

The Law of Portfolio Learnability

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I. Introduction

In the Markowitz (1952) framework, first the portfolio manager estimate the two moments of returns, secondly the PM estimate the portofflio wieghts optimising a criterion.

A different view emerged with the direct portfolio approach of Brandt (1999) and Brandt et al. (2009). Investors ultimately care about portfolio weights, not forecasts of moments. Rather than estimating conditional expected returns and covariance matrices as intermediate objects, one can parameterize and estimate the portfolio policy directly. This changed the object of estimation. This was a very futuristic approach, anticipating of years what computer science and machine learning is doing now.

This estimation problem becomes particularly acute in high-dimensional finance. Kozak et al. (2020) show that the estimation becomes unstable in low-eigenvalue directions. Where the eigenvalues tend to zero, the payoff of the eigenportfolio tend to infinite, creating an arbitrage opportunity. This motivate the economically shrinkage.

More recently, the machine-learning literature has brought model complexity to the foreground. Empirical asset pricing means model with data the financial market. There is always a model behind every modelization. Thus finance can't ignore what is happening in the machine leanring world. Indeed Kelly et al. (2024) and Kelly and Malamud (2025) show that increasing model complexity can improve out-of-sample return prediction even deep into the overparameterized regime. Their conclusion is deliberately strong:

"The analyst should always use the largest approximating model that she can compute."

This was strongly new for the finance world in which the only virute was the parsimony.

This "virtue of complexity" shows that overparameterization can improve approximation and generate useful implicit regularization. Nagel (2025), in turn, shows why the economic meaning of highly overparameterized estimators must be interpreted carefully: large random-feature models can converge to kernel rules whose behavior is driven by particular forms of historical weighting

rather than by an arbitrarily rich set of distinct economic decisions, introducing nominally and the effective complexity.

The overparametrized model not only now are in returns prediction, but appears in directly portfolio choice. Simon et al. (2026) replace the linear parametric portfolio policy with deep neural networks and show that flexible nonlinear policies can generate economically valuable allocations, while investor preferences themselves act as a form of economic regularization.

Taken together, this literature establishes that rich representations can be useful and that regularization is if not indispensable, at least useful.

Yet several fundamental questions remain open.

Building on the direct portfolio approach of Brandt: How reliably, and how quickly, can a flexible and "complex" portfolio rule recover the population-optimal investment decision?

Building on the spectral shrinkage argument of Kozak, Nagel, and Santosh: Which investment directions should be shrunk because they are only weakly identified by returns?

Building on the virtue-of-complexity literature: When is a richer representation a virtue for portfolio choice, and what notion of complexity is economically relevant for the investment decision itself?

Answering these questions requires starting from a more fundamental one:

How much of the optimal portfolio can be learned from a finite market history?

Answering this question requires establishing whether and how fast the portfolio decision can be learned directly, identifying which investment directions should be shrunk because they are only weakly revealed by realized returns, and determining what notion of complexity is economically relevant for portfolio choice—and how much of that complexity a finite market history can support.

First we used a particular estimator (Kernel ridge regression) to quantify our results, later we generalize these to every estimator. Our first contribution is to quantify how rapidly direct portfolio

estimation recovers the population tangency direction as market history accumulates. We prove that the directly weights estimation for maximizing the Sharpe ratio is consistent and we characterize the rate at which its estimated direction converges to the population Sharpe. We find that portfolio learnability is governed by the spectrum of managed portfolios. Directions associated with large eigenvalues correspond to investment opportunities that generate substantial payoff variation and are therefore well identified from the available return history. By contrast, low-eigenvalue directions are only weakly identified, so taking large positions along them makes the portfolio especially sensitive to estimation error. More available portfolio directions therefore do not necessarily translate into more usable investment opportunities: what matters is how much of the opportunity set is concentrated in directions that can be estimated with sufficient precision. The result therefore provides both a formal justification for direct portfolio estimation and a quantitative answer to how much market history is required to learn the optimal investment decision.

Our first contribution implies even a quantification of the optimal shrinkage: It should decline as market history accumulates and as a function of the managed portfolio spectrum. Additional observations progressively reveal portfolio directions that were previously too weakly identified to support reliable investment positions, allowing the investor to exploit a broader set of economically meaningful opportunities.

Our second contribution is what we call the law of effective portfolio complexity. We distinguish two notions of complexity that play fundamentally different economic roles. The first is nominal representation complexity: the size of the approximating representation, as measured, for example, by the number of features, basis functions, or parameters. Kelly, Malamud, and Zhou show that, under their proportional high-dimensional asymptotic framework, increasing nominal representation size can improve limiting out-of-sample return-prediction performance. Richer representations can therefore be valuable because they expand approximation capacity and the set of investment rules that can potentially be expressed.

The second notion is effective portfolio complexity, which we define directly in the space of the investment decision. It measures the effective breadth of the portfolio: the effective number of

economically distinct managed-portfolio directions that remain active after shrinkage. This quantity need not move one-for-one with representation size. A representation may be arbitrarily rich while the portfolio learned from it remains effectively low dimensional. We show that the optimal level of effective portfolio complexity is finite at every finite sample size and increases as additional market history becomes available. Too little effective complexity leaves valuable investment opportunities unused; moving beyond the data-supported scale increasingly activates directions that the available history only weakly identifies. A rich representation may be a virtue; how much of that richness should become an active investment decision is a separate learnability question.

Our third and last contribution is what we call the fundamental limit of portfolio learning. We show that the rate at which the optimal portfolio can be recovered is not merely a property of our estimator. No portfolio-learning procedure—including neural networks, random forests, or alternative kernel methods—can uniformly improve this rate over the same class of economies without imposing additional structure.

The learning limit therefore does not belong to a particular estimator. It belongs to the investment problem itself.

Taken together, our results characterize how rapidly the population tangency portfolio can be recovered from finite market history, how richer representations expand the set of potential investment opportunities, and how the effective portfolio complexity that can be reliably exploited grows as additional data become available. We also establish the fastest rate at which any procedure can achieve this recovery. Our results therefore quantify both the opportunities created by increasingly flexible representations and the fundamental limits that finite market history imposes on machine learning for portfolio choice in finance.

II. Population Problem Portfolio Choice

At date t , the portfolio manager faces N risky stocks. The corresponding next-period excess-return vector is $R_{t+1} = (R_{1,t+1}, \dots, R_{N,t+1})' \in \mathbb{R}^N$. For each stock $i = 1, \dots, N$, the portfolio manager observes a D -dimensional vector of firm characteristics $Z_{i,t} \in \mathcal{Z} \subseteq \mathbb{R}^D$. The portfolio manager has to choose the portfolio weights policy, between an infinite number of policies in an Hilbert space $\mathcal{H}$, at every time, this maps characteristics into weights: $W \in \mathcal{H} : \mathcal{Z} \rightarrow \mathbb{R}$. The resulting portfolio return at each time is

$$R_{t+1}^p(W) = \frac{1}{N} \sum_{i=1}^N W(Z_{i,t}) R_{i,t+1}. \quad (1)$$

The factor $1/N$ is a cross-sectional normalization that prevents portfolio scale from growing mechanically with the number of available stocks.

Definition 1 (Economy). An economy ε is a probability law for the stochastic process

$$\{(N, Z_t, R_{t+1})\}_{t \in \mathbb{Z}},$$

defined on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in \mathbb{Z}}, \mathbb{P}_\varepsilon)$. We require Z_t to be $\mathcal{F}_t$ -measurable and R_{t+1} to be $\mathcal{F}_{t+1}$ -measurable.

For a fixed economy, expectations are taken under $\mathbb{P}_\varepsilon$; we suppress the economy subscript when it is unambiguous. The population criterion is

$$Q(W) = \mathbb{E} \left[\left(1 - R_{t+1}^p(W) \right)^2 \right], \quad W^\star \in \arg \min_{W \in \mathcal{H}} Q(W). \quad (2)$$

The following proposition relates this criterion to the maximum squared Sharpe ratio. All population statements refer to the same one-period state-return distribution.

Proposition 1 (Maximum-Sharpe direction; cf. Britten-Jones (1999)). *Suppose every policy in $\mathcal{H}$ generates a square-integrable payoff, the population minimum is attained, and $0 < Q(W^\star) < 1$. Sharpe ratios below are evaluated only for policies with positive payoff variance. Every nonzero policy $W \in \mathcal{H}$ can be written as*

$$W = cU, \quad c \in \mathbb{R}, \quad \|U\|_{\mathcal{H}} = 1.$$

The normalized policy U determines the portfolio direction, namely the relative state-dependent allocation rule, whereas c determines the overall scale of exposure along that direction. Then, with the minimization over U restricted to positive-variance payoffs,

$$\min_{W \in \mathcal{H}} Q(W) = \min_{\substack{U \in \mathcal{H} \\ \|U\|_{\mathcal{H}}=1}} \frac{1}{1 + \text{SR}^2(R_{t+1}^p(U))},$$

where

$$\text{SR}(R_{t+1}^p(U)) := \frac{\mathbb{E}[R_{t+1}^p(U)]}{\sqrt{\text{Var}(R_{t+1}^p(U))}}.$$

Hence, minimizing Q selects a maximum-Sharpe direction and fixes its scale, with positive expected excess return.

We embrace the philosophy of Brandt (1999): rather than estimating expected returns and the covariance matrix as intermediate objects and then solving for the optimal portfolio, we estimate portfolio weights directly. Estimating the mean return vector requires estimating N expected returns, while estimating the covariance matrix requires estimating $N(N + 1)/2$ distinct covariance parameters. By contrast, directly estimating the portfolio means estimating the N portfolio weights that are ultimately the object of interest.

The criterion in (2) deliberately describes a frictionless portfolio problem. More generally, the loss should represent the economic objective actually faced by the portfolio manager. Transaction

costs, market impact, short-selling costs, leverage constraints, or other implementation frictions should therefore be incorporated directly into the portfolio objective whenever they are economically relevant. Estimating the frictionless-optimal policy and subtracting costs afterwards generally does not deliver the same decision.

Proposition 2 (Pricing kernel and Hansen–Jagannathan frontier). *Let $W_\varepsilon^\star$ denote the population-optimal policy in Proposition 1, and define*

$$m_{t+1,\varepsilon}^\star := 1 - R_{t+1}^p(W_\varepsilon^\star).$$

Then $m_{t+1,\varepsilon}^\star$ prices every managed excess-return portfolio:

$$\mathbb{E}_\varepsilon[m_{t+1,\varepsilon}^\star R_{t+1}^p(V)] = 0, \quad \forall V \in \mathcal{H}.$$

Moreover, letting $\text{SR}_\varepsilon^\star := \text{SR}(R_{t+1}^p(W_\varepsilon^\star))$,

$$\mathbb{E}_\varepsilon[m_{t+1,\varepsilon}^\star] = \mathbb{E}_\varepsilon[(m_{t+1,\varepsilon}^\star)^2] = \frac{1}{1 + (\text{SR}_\varepsilon^\star)^2}.$$

Hence the normalized pricing kernel

$$\tilde{m}_{t+1,\varepsilon} := \frac{m_{t+1,\varepsilon}^\star}{\mathbb{E}_\varepsilon[m_{t+1,\varepsilon}^\star]}$$

satisfies $\mathbb{E}_\varepsilon[\tilde{m}_{t+1,\varepsilon}] = 1$, and $\text{Var}_\varepsilon(\tilde{m}_{t+1,\varepsilon}) = (\text{SR}_\varepsilon^\star)^2$. Equivalently,

$$\frac{\sigma_\varepsilon(\tilde{m}_{t+1,\varepsilon})}{\mathbb{E}_\varepsilon[\tilde{m}_{t+1,\varepsilon}]} = \text{SR}_\varepsilon^\star,$$

so the Hansen–Jagannathan bound ((Hansen and Jagannathan, 1991)) holds with equality.

The proposition provides the asset-pricing dual of Proposition 1. The same portfolio that maximizes the Sharpe ratio generates, through $m_{t+1,\varepsilon}^\star = 1 - R_{t+1}^p(W_\varepsilon^\star)$, the minimum-variance

unit-mean pricing kernel for the managed-payoff space. Thus the maximum-Sharpe portfolio and the Hansen–Jagannathan frontier are two representations of the same population geometry.

III. Direct Portfolio Learning

To estimate the portfolio policy we use a kernel representation Filipović and Schneider (2024); Nagel (2025) because it provides a common framework for portfolio policies of very different forms, as shown in table Indeed, depending on the kernel, the same estimator can generate static, linear, polynomial or sieve-based, and flexible nonlinear mappings, generalizing respectively the static mean–variance portfolio of Britten-Jones (1999), the linear parametric portfolio policies of Brandt et al. (2009), and the semiparametric spline-based policies of Caldeira et al. (2023).

Kernel methods are also closely connected to the recent work of Nagel (2025); Kelly and Malamud (2025). In particular, random Fourier features (Rahimi and Recht, 2007) provide finite-dimensional approximations to stationary kernels, while the corresponding kernel estimator can be viewed as the infinite-feature limit. In this sense, the kernel formulation provides a natural benchmark for portfolio models whose nominal representation complexity is allowed to become arbitrarily large.

Kernel methods are also closely related to neural-network representations. A random-feature approximation can be interpreted as a neural network with a fixed, randomly generated hidden layer, in which only the final linear coefficients are estimated. More generally, neural tangent kernels arise as the infinite-width limits of particular neural-network training regimes (Jacot et al., 2018).

Let

$$K : \mathcal{Z} \times \mathcal{Z} \rightarrow \mathbb{R}$$

be a positive-definite scalar kernel and let $\mathcal{H}_K$ denote the corresponding space of characteristic-based portfolio policies.

Assumption 1 (Portfolio representation). The space $\mathcal{Z}$ is Polish. Fix a jointly measurable kernel K , with separable RKHS $\mathcal{H}_K$, satisfying

$$\sup_{z \in \mathcal{Z}} K(z, z) \leq \kappa_K < \infty.$$

The representation is fixed across economies and sample sizes.

Let $W_K^\star$ denote the minimum-norm population minimizer in $\mathcal{H}_K$, when attained. Write $W_\varepsilon^\star = W_{\varepsilon,K}^\star$ and $\Sigma_{\mathcal{H}_K,\varepsilon}$ when distinguishing economies.

The empirical regularized counterpart of the population problem in Equation (2) is

$$\begin{aligned} \widehat{W}_{\lambda,T} &:= \arg \min_{W \in \mathcal{H}_K} \left\{ \frac{1}{T} \sum_{t=1}^T \left(1 - \frac{1}{N} \sum_{i=1}^N W(Z_{i,t}) R_{i,t+1} \right)^2 + \lambda \|W\|_{\mathcal{H}_K}^2 \right\} \\ &= \arg \min_{W \in \mathcal{H}_K} \left\{ \widehat{Q}(W) + \lambda \|W\|_{\mathcal{H}_K}^2 \right\} = \min_{\substack{U \in \mathcal{H} \\ \|U\|_{\mathcal{H}}=1}} \left\{ \frac{1}{1 + \widehat{\mathbf{SR}}_{\lambda,T}^2(R_{t+1}^p(U))} \right\}, \quad \lambda > 0. \end{aligned} \tag{3}$$

where

$$\widehat{\mathbf{SR}}_{\lambda,T}(R_{t+1}^p(U)) := \frac{\frac{1}{T} \sum_{t=1}^T R_{t+1}^p(U)}{\sqrt{\frac{1}{T} \sum_{t=1}^T \left(R_{t+1}^p(U) - \frac{1}{T} \sum_{s=1}^T R_{s+1}^p(U) \right)^2 + \lambda}}.$$

The economic structure of the estimator becomes transparent by viewing kernel-managed portfolio payoffs as elements of the portfolio policy space. At each date t , define

$$X_t := \frac{1}{N} \sum_{i=1}^N R_{i,t+1} K(\cdot, Z_{i,t}) \in \mathcal{H}_K.$$

Thus X_t is a random element of $\mathcal{H}_K$. If evaluated at $z \in \mathcal{Z}$, it gives

$$X_t(z) = \frac{1}{N} \sum_{i=1}^N R_{i,t+1} K(z, Z_{i,t}) \in \mathbb{R}.$$

the payoff generated by the kernel-managed portfolio that assigns stock i the weight $K(z, Z_{i,t})/N$.

Its population mean is

$$\mu_{\mathcal{H}_K} := \mathbb{E}[X_t] \in \mathcal{H}_K,$$

while its population second-moment operator is

$$\Sigma_{\mathcal{H}_K} := \mathbb{E}[X_t \otimes X_t] : \mathcal{H}_K \rightarrow \mathcal{H}_K,$$

where, for every $V \in \mathcal{H}_K$,

$$(X_t \otimes X_t)V = \langle V, X_t \rangle_{\mathcal{H}_K} X_t.$$

The first moment captures expected managed-portfolio payoffs, while the second-moment operator describes the magnitude of realized payoffs across portfolio directions.

Their sample counterparts are the element

$$\hat{\mu}_T = \frac{1}{T} \sum_{t=1}^T X_t \in \mathcal{H}_K,$$

and the operator

$$\hat{\Sigma}_T = \frac{1}{T} \sum_{t=1}^T X_t \otimes X_t : \mathcal{H}_K \rightarrow \mathcal{H}_K.$$

Proposition 3 (Managed-portfolio Markowitz representation). *The population-optimal minimum-norm policy satisfies*

$$\mu_{\mathcal{H}_K} = \Sigma_{\mathcal{H}_K} W_K^\star. \quad (4)$$

Equivalently,

$$W_K^\star = \Sigma_{\mathcal{H}_K}^\dagger \mu_{\mathcal{H}_K},$$

where $\Sigma_{\mathcal{H}_K}^\dagger$ denotes the pseudo inverse.

The regularized sample policy in Equation (3) satisfies

$$\hat{W}_{\lambda,T} = \left(\hat{\Sigma}_T + \lambda I \right)^{-1} \hat{\mu}_T, \quad \lambda > 0. \quad (5)$$

Equation (5) provides the structural link to classical portfolio choice. The investor is still

combining expected returns against second moments, as in Markowitz. The difference is that the investment objects are now characteristic-managed portfolios weighted with the kernels.

Regularization has a familiar portfolio interpretation. As emphasized by Kozak et al. (2020), portfolio directions with very little payoff variation but non-negligible expected returns resemble near-arbitrage opportunities and should therefore be treated with particular caution. Regularization has a familiar portfolio interpretation. As emphasized by Kozak et al. (2020), weak portfolio directions are particularly dangerous because small estimation errors can translate into very large portfolio positions. In their eigenportfolio setting, a low-variance direction with a sufficiently large estimated expected return can even resemble a near-arbitrage opportunity.

The same instability appears in our managed-portfolio representation. Let $\Sigma_{\mathcal{H}_K} e_j = \mu_j e_j$, and let $m_j := \langle \mu_{\mathcal{H}_K}, e_j \rangle_{\mathcal{H}_K}$. Without regularization, the portfolio weight along direction e_j is proportional to

$$\widehat{W}_{0,j} = \frac{\widehat{m}_j}{\widehat{\mu}_j}.$$

Hence, when $\widehat{\mu}_j$ is small, even modest estimation error in $\widehat{m}_j$ can generate a very large portfolio position. Regularization replaces the small eigenvalue in the denominator by $\widehat{\mu}_j + \lambda$:

$$\widehat{W}_{\lambda,j} = \frac{\widehat{m}_j}{\widehat{\mu}_j + \lambda} = \frac{\widehat{\mu}_j}{\widehat{\mu}_j + \lambda} \widehat{W}_{0,j}.$$

Thus, $\widehat{\mu}_j / (\widehat{\mu}_j + \lambda)$ is the spectral shrinkage factor.

The spectral filter derived above suggests a natural measure of how much of the managed-portfolio opportunity set is effectively used by the investor. The relevant notion of portfolio complexity is not the number of characteristics, parameters, or random features in the representation, but the number of economically distinct portfolio directions that remain active after shrinkage.

This motivates the following definition.

Definition 2 (Effective portfolio complexity). For a fixed portfolio representation K and $\lambda > 0$,

define

$$C_K(\lambda) := \text{Tr}[\Sigma_{\mathcal{H}_K}(\Sigma_{\mathcal{H}_K} + \lambda I)^{-1}] = \sum_{j \geq 1} \frac{\mu_j}{\mu_j + \lambda}. \quad (6)$$

Effective portfolio complexity is the shrinkage-adjusted number of managed-portfolio directions used by the portfolio rule.

The definition follows directly from the spectral shrinkage mechanism already introduced. Each managed-portfolio direction contributes between zero and one to $C_K(\lambda)$. If $\mu_j \gg \lambda$, its shrinkage factor is close to one and the direction is effectively active. If $\mu_j \ll \lambda$, the direction contributes little and is effectively suppressed.

Effective complexity therefore measures how deeply into the managed-portfolio spectrum the investor is actually operating.

This notion is fundamentally different from nominal representation size. A model may contain thousands, or infinitely many, basis directions while the portfolio decision itself remains effectively low dimensional because regularization suppresses most weak directions. Conversely, a rich representation can be valuable because it makes additional investment opportunities available once sufficient market history exists to identify them.

IV. Portfolio Learnability

A. A Class of Portfolio Economies

For the fixed representation in Assumption 1, fix $b > 1$, $\gamma > b$, constants $C_\beta, B_X, R > 0$, $K_4 \geq 1$, and $0 < c_\mu \leq C_\mu < \infty$. We denote by

$$\mathcal{E} = \mathcal{E}(b, \gamma, C_\beta, B_X, K_4, R, c_\mu, C_\mu)$$

the nonempty class of economies $\mathcal{E}$ satisfying the following conditions, with all bounds common to the class:

(E1) The process $\{(Z_t, R_{t+1})\}_{t \in \mathbb{Z}}$ is strictly stationary and polynomially β -mixing, with $\beta_\varepsilon(k) \leq C_\beta k^{-\gamma}$ for every $k \geq 1$.

(E2) *Gaussian-type managed-return tails and conditional fourth moments.* Let

$$K_t := \left[K(Z_{i,t}, Z_{j,t}) \right]_{i,j=1}^N.$$

Assume

$$\mathbb{E}_\varepsilon \left[\exp \left(\frac{N^{-2} R'_{t+1} K_t R_{t+1}}{B_X^2} \right) \middle| \mathcal{F}_t \right] \leq 2 \quad \text{a.s.} \quad (7)$$

For every $V \in \mathcal{H}_K$, assume also

$$\mathbb{E}_\varepsilon \left[(R_{t+1}^p(V))^4 \middle| \mathcal{F}_t \right] \leq K_4 \left\{ \mathbb{E}_\varepsilon \left[(R_{t+1}^p(V))^2 \middle| \mathcal{F}_t \right] \right\}^2 \quad \text{a.s.} \quad (8)$$

Both bounds are uniform over the class. Since $N^{-2} R'_{t+1} K_t R_{t+1} = \|X_t\|_{\mathcal{H}_K}^2$, Condition (7) directly controls the diagonal of the induced kernel while allowing unbounded returns. Conditional Gaussian returns with uniformly bounded conditional second moments satisfy Condition (E2); see Proposition 4.

(E3) The population minimum over $\mathcal{H}_K$ is attained. Its minimum-norm solution $W_\varepsilon^\star$ satisfies $\|W_\varepsilon^\star\|_{\mathcal{H}_K} \leq R$.

(E4) The pricing kernel $m_{t+1,\varepsilon}^\star := 1 - R_{t+1}^p(W_\varepsilon^\star)$ satisfies

$$\mathbb{E}_\varepsilon[m_{t+1,\varepsilon}^\star R_{t+1}^p(V) \mid \mathcal{F}_t] = 0, \quad \forall V \in \mathcal{H}_K, \quad \text{a.s.}$$

Here $\mathcal{F}_t$ is the investor's information at the decision date in Definition 1. Conditional pricing is imposed separately from unconditional population optimality; positivity is not imposed.

(E5) Let $\Sigma_{\mathcal{H}_K,\varepsilon} = \mathbb{E}_\varepsilon[X_t \otimes X_t]$ and let $\mu_{1,\varepsilon} \geq \mu_{2,\varepsilon} \geq \dots > 0$ denote its positive eigenvalues. Then

$$c_\mu j^{-b} \leq \mu_{j,\varepsilon} \leq C_\mu j^{-b}, \quad j \geq 1.$$

(E6) The optimal payoff has positive variance and $\text{SR}_{\varepsilon,K}^\star := \text{SR}(R_{t+1}^p(W_\varepsilon^\star))$ satisfies the common bounds

$$0 < \underline{S} \leq \text{SR}_{\varepsilon,K}^\star \leq \bar{S} < \infty.$$

Condition (E2) replaces the almost-sure bounded-return restriction by Gaussian-compatible tail and fourth-moment control. Proposition 4 in Appendix A.4 gives a simple sufficient conditional-Gaussian specification. No Gaussianity is otherwise imposed.

The class $\mathcal{E}$ separates two forces governing portfolio learnability. The exponent γ controls the persistence of the data, while b controls the geometry of the managed-portfolio spectrum. The parameter b describes how concentrated the managed-portfolio opportunity set is. A large b means that most payoff variation is concentrated in a relatively small number of strong portfolio directions. A smaller b leaves a longer tail of economically distinct but progressively weaker directions.

Polynomial spectral decay is natural for finite-smoothness representations, including Sobolev–Matérn, Laplace, and spline-type kernels. Gaussian kernels can instead generate exponential or faster-than-polynomial decay (Belkin, 2018). For exposition, we use $b = \infty$ as shorthand for the

fast-decay benchmark, including Gaussian-type spectra and the finite-rank limit. The class $\mathcal{E}$ itself has finite b ; these are separate benchmarks, not substitutions into the restriction $\gamma > b$. Their relevance depends on the spectrum of $\Sigma_{\mathcal{H}_K, \varepsilon}$, not on the kernel name alone.

B. Learnable Sharpe ratio and optimal shrinkage

We can now ask how regularization should change as market history accumulates, how rapidly the population maximum-Sharpe direction can be recovered, and how much of the managed-portfolio opportunity set can be reliably used. The financial observations can be independent over time or weakly dependent.

Population loss and Sharpe of an estimated policy hold the training sample fixed and integrate over the stationary one-period evaluation law. They are not the Sharpe ratio of a finite test sample.

Theorem 1 (Optimal shrinkage and learnable frontier). *Choose the optimal regularization*

$$\lambda_T^* \asymp T^{-b/(b+1)}$$

, with comparison constants independent of the economy. Uniformly over $\varepsilon \in \mathcal{E}$,

$$Q_\varepsilon(\widehat{W}_{\lambda_T^*, T}) - Q_\varepsilon(W_\varepsilon^*) = O_{\mathbb{P}_\varepsilon}\left(T^{-b/(b+1)}\right). \quad (9)$$

Moreover, uniformly over $\varepsilon \in \mathcal{E}$,

$$\text{SR}_{\varepsilon, K}^* - \text{SR}\left(R_{t+1}^p(\widehat{W}_{\lambda_T^*, T})\right) = O_{\mathbb{P}_\varepsilon}\left(T^{-b/(b+1)}\right), \quad (10)$$

where:

$$SR_{\varepsilon}(\widehat{W}_{K,\lambda,T}) = \frac{\mathbb{E}_{\varepsilon} \left[R_{t+1}^p(\widehat{W}_{K,\lambda,T}) \middle| \widehat{W}_{K,\lambda,T} \right]}{\sqrt{\text{Var}_{\varepsilon} \left(R_{t+1}^p(\widehat{W}_{K,\lambda,T}) \middle| \widehat{W}_{K,\lambda,T} \right)}} = \frac{\mathbb{E}_{\varepsilon} \left[\frac{1}{N} \sum_{i=1}^N \widehat{W}_{K,\lambda,T}(Z_{i,t}) R_{i,t+1} \middle| \widehat{W}_{K,\lambda,T} \right]}{\sqrt{\text{Var}_{\varepsilon} \left(\frac{1}{N} \sum_{i=1}^N \widehat{W}_{K,\lambda,T}(Z_{i,t}) R_{i,t+1} \middle| \widehat{W}_{K,\lambda,T} \right)}}$$

and

$$SR_{\varepsilon,K}^{\star} = \sup_{W \in \mathcal{H}_K} \frac{\mathbb{E}_{\varepsilon} \left[\frac{1}{N} \sum_{i=1}^N W(Z_{i,t}) R_{i,t+1} \right]}{\sqrt{\text{Var}_{\varepsilon} \left(\frac{1}{N} \sum_{i=1}^N W(Z_{i,t}) R_{i,t+1} \right)}}.$$

The theorem first establishes consistency of direct portfolio learning. Since $\lambda_T^{\star} \rightarrow 0$ and $SR_K^{\star} - SR(\widehat{W}_{\lambda_T^{\star},T}) \rightarrow_P 0$, the portfolio manager eventually recovers the maximum-Sharpe direction available within the chosen portfolio representation.

The optimal regularization path also gives a dynamic interpretation to the eigenportfolio logic discussed above. With a short return history, directions deep in the managed-portfolio spectrum are weakly identified and are therefore shrunk heavily. As more observations become available, their expected payoff and payoff variation can be estimated more precisely, allowing the manager to take progressively larger positions in those directions.

Indeed, as $\lambda_T^{\star} \asymp T^{-b/(b+1)}$ falls with T , the spectral filter $\frac{\mu_j}{\mu_j + \lambda_T^{\star}}$ moves toward one for progressively smaller values of μ_j . Directions that were effectively suppressed in short samples therefore become statistically usable in longer samples.

Regularization is thus a finite-history device. More data allow the portfolio manager to move deeper into the managed-portfolio spectrum and exploit investment opportunities that were previously too weakly identified.

The shape of the spectrum determines how quickly portfolio opportunities can be learned. A steep spectrum (large b) concentrates economically relevant payoff variation in a few directions, making the portfolio easier to learn. A flatter spectrum spreads variation across many weaker directions, increasing potential breadth but slowing learning.

Importantly, portfolio learnability is therefore not determined directly by the number D of firm characteristics. Two representations with the same number of characteristics can generate very different learning problems. If their characteristics produce largely redundant portfolio bets, the managed-portfolio spectrum is concentrated. If instead they produce many distinct but weak portfolio directions, the spectrum is diffuse and substantially more market history is required.

What matters for the portfolio manager is therefore not the raw number of signals, but the economic geometry of the portfolios generated by those signals.

Figure 1 gives a graphical interpretation of the learning result: as market history grows, the learned capital-allocation line rotates toward the population tangency line. Under Condition (E6), at any fixed volatility, the expected-return gap closes at rate $O_P\left(T^{-b/(b+1)}\right)$.

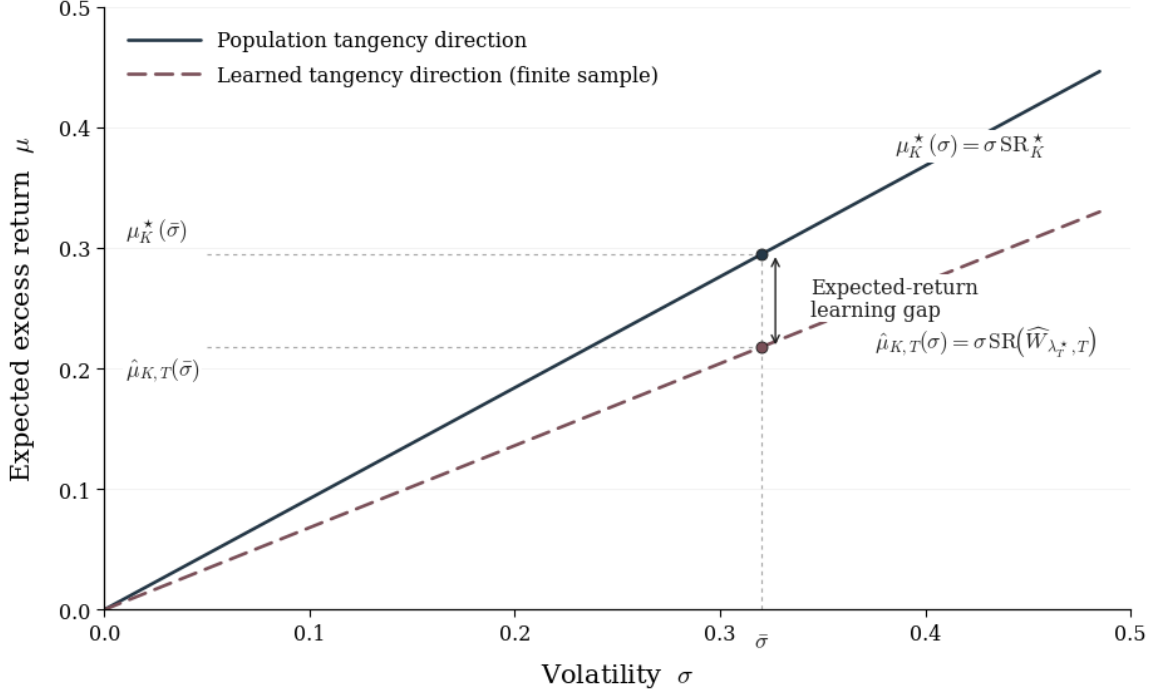


Figure 1. Learning the tangency direction. The population tangency direction generates expected excess return $\mu_K^*(\sigma) = \sigma \text{SR}_K^*$, while the direction learned from a finite history generates $\hat{\mu}_{K,T}(\sigma) = \sigma \text{SR}(\hat{W}_{\lambda_T^*, T})$. At the fixed volatility target $\bar{\sigma}$, the vertical distance is the expected-return learning gap $\mu_K^*(\bar{\sigma}) - \hat{\mu}_{K,T}(\bar{\sigma})$.

C. Optimal effective portfolio complexity

The optimal-shrinkage result can now be expressed directly in terms of portfolio complexity.

Theorem 2 (Learnable effective portfolio complexity). *Under the portfolio-loss conditions of Theorem 1, the rate-optimal effective complexity satisfies*

$$C_{K,T}^* := C_K(\lambda_T^*) \asymp T^{1/(b+1)}. \quad (11)$$

The theorem gives a direct economic interpretation of additional market history. As T increases and optimal shrinkage declines, the portfolio manager can reliably activate a broader set of managed-portfolio directions.

This expansion is nevertheless gradual. Effective complexity grows sublinearly: $\frac{C_{K,T}^*}{T} \rightarrow 0$.

Thus, even when the underlying representation is arbitrarily rich, a finite market history supports only a limited fraction of its potential portfolio directions.

A steep managed-portfolio spectrum implies slower growth of effective complexity because most economically relevant payoff variation is already concentrated in a few strong directions. A flatter spectrum supports greater potential breadth, but exploiting that breadth requires substantially more market history.

Figure 2 summarizes this tradeoff. With a short history, the Sharpe-maximizing portfolio uses only a small number of effective directions. With more observations, weaker directions become reliably investable and the optimal effective complexity shifts outward.

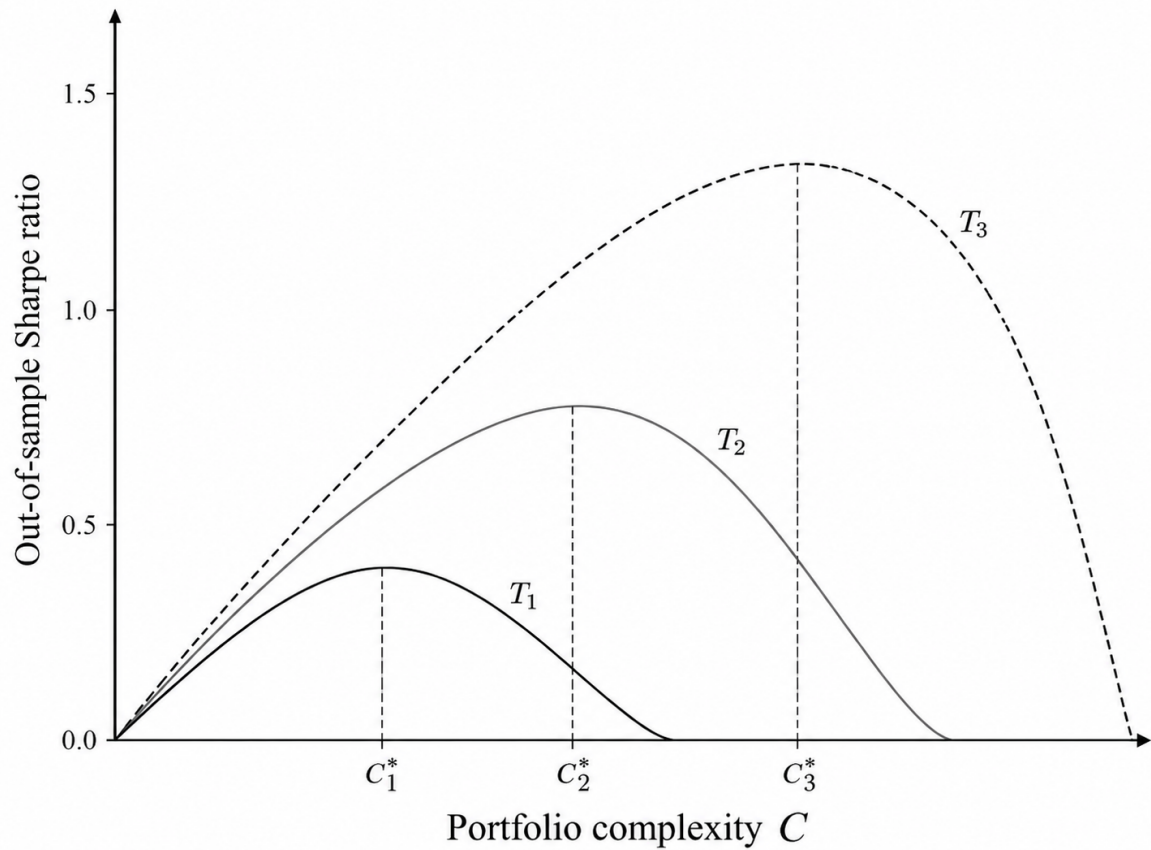


Figure 2. Market history and learnable portfolio complexity. Schematic Sharpe–complexity profiles for different history lengths. Longer market histories allow the portfolio manager to activate a broader set of managed-portfolio directions, shifting the Sharpe-maximizing effective complexity to the right.

The distinction from nominal size is central. A large representation expands the set of portfolio policies that can potentially be expressed. Effective complexity measures how much of that representation the portfolio manager can actually use given the available history of returns. Effective complexity is therefore a learnable economic that quantify the amount of eigenportfolios that the portfolio manager trade.

D. The fundamental limit

The previous results describe how a portfolio manager should use a finite market history: weak managed-portfolio directions should be shrunk, the amount of shrinkage should decline as more observations become available, and the set of portfolio opportunities that can be reliably exploited expands with the sample. A natural question remains. Could a different learning technology overcome this limitation? Could a sufficiently large neural network, a richer random-feature model, or a more sophisticated portfolio algorithm recover the optimal investment decision substantially faster?

The answer is no without additional economic structure.

Theorem 3 (Fundamental limit of maximum-Sharpe learning). *There exists a constant $c > 0$ such that, for all sufficiently large T ,*

$$\inf_{\tilde{W}_T} \sup_{\varepsilon \in \mathcal{E}} \mathbb{E}_{\varepsilon} \left[\text{SR}_{\varepsilon, K}^{\star} - \text{SR}_{\varepsilon} \left(R_{t+1}^p(\tilde{W}_T) \right) \right] \geq c T^{-b/(b+1)}.$$

The infimum is over all measurable portfolio-learning procedures based on the T observed financial rows. Hence no portfolio-learning procedure can uniformly recover the population maximum Sharpe ratio at a faster rate than $T^{-b/(b+1)}$ over $\mathcal{E}$.

Together with Theorem 1, this establishes $T^{-b/(b+1)}$ as the minimax-optimal rate for learning the maximum Sharpe ratio over the class considered here.

The limitation is therefore informational rather than computational. Portfolio directions deep in the managed-payoff spectrum are only weakly identified by a finite history of realized returns. A richer model may represent more investment opportunities, and a better algorithm may use the available information more effectively, but neither can make those weakly identified opportunities more visible in the data.

Learning faster requires knowing more about the economy. This can come from a longer market history or from additional economic restrictions on the return process, the managed-payoff spectrum,

or the location of optimal portfolio demand. Absent such information, the learning rate is a property of the investment problem, not of the particular algorithm used to solve it.

V. Representation, Learnability, and the Virtue of Complexity

Kelly et al. (2024) study an environment in which expanding the model also captures an increasing share of the underlying economic signal. In their setting, the improvement in approximation can dominate the additional estimation cost. This is the virtue of complexity.

Our question is different. We hold the investor's information set fixed and vary only the richness of the portfolio representation. We take the view that a portfolio manager should make available all information she considers economically useful and can observe in real time. Regularization, rather than ex ante feature exclusion, should then determine how much of the resulting portfolio opportunity set can actually be learned from the available data. We therefore study the virtue of Complexity, as the virtue of richer representation, separately from the addition of new signals.

Our point in this section is that a richer portfolio representation and a better learned portfolio are two different objects.

Let SR^* denote the maximum population Sharpe ratio over the space of L^2 portfolio policies,

$$SR^* := \sup_{W \in L^2} SR(W),$$

and let

$$SR_K^* := \sup_{W \in \mathcal{H}_K} SR(W)$$

denote the maximum population Sharpe ratio available within representation K , where $\mathcal{H}_K \subseteq L^2$ is the corresponding policy space. For example, a linear kernel restricts portfolio weights to be linear functions of characteristics, whereas a Matérn kernel allows a richer nonlinear class.

Finally,

$$SR(\widehat{W}_{K,\lambda,T})$$

denotes the population Sharpe ratio of the portfolio policy learned within representation K from a market history of length T .

The gap between the unrestricted population opportunity set and the portfolio actually learned from the data can be decomposed as

$$\text{SR}^\star - \text{SR}(\widehat{W}_{K,\lambda,T}) = \underbrace{\text{SR}^\star - \text{SR}_K^\star}_{\text{representation gap}} + \underbrace{\text{SR}_K^\star - \text{SR}(\widehat{W}_{K,\lambda,T})}_{\text{learning gap}}. \quad (12)$$

The representation gap measures the investment opportunities that cannot be expressed by the chosen representation, even if the investment environment were known perfectly. The learning gap measures how much of the opportunity set available within that representation cannot be recovered from a finite history of returns.

Now consider two nested representations,

$$\mathcal{H}_{K_1} \subseteq \mathcal{H}_{K_2}.$$

At the population level,

$$\text{SR}_{K_2}^\star \geq \text{SR}_{K_1}^\star.$$

A richer representation can therefore only expand the population investment opportunity set. In this precise sense, complexity is a virtue for representation.

But this population ordering need not survive estimation from finite data. It is perfectly possible that

$$\text{SR}_{K_2}^\star > \text{SR}_{K_1}^\star$$

while

$$\text{SR}(\widehat{W}_{K_2,\lambda_2,T}) < \text{SR}(\widehat{W}_{K_1,\lambda_1,T}).$$

The larger representation contains better population opportunities, but some of those opportunities may lie in managed-portfolio directions that are too weakly identified by the available return history.

A richer representation therefore expands what the portfolio manager could exploit. Whether

those additional opportunities improve the learned portfolio depends on whether they can actually be learned from the available data.

VI. Simulation: a balanced three-factor economy

We study direct portfolio learning in a stock-level economy with three economic factors. The managed spectrum is generated by the characteristic distribution and the Matérn kernel. Numerical kernel rank is distinct from the number of economic factors. Table I gives the frozen calibration.

A. Characteristics, returns, and the population policy

There are $N = 3G$ stocks and $D = 6$ characteristics. Independent group-level states start from $S_{g,d,0} \sim N(0, 1)$ and evolve as

$$S_{g,d,t+1} = \rho_d S_{g,d,t} + \sqrt{1 - \rho_d^2} \eta_{g,d,t+1}, \quad \eta_{g,d,t+1} \sim N(0, 1).$$

With $U_{g,d,t} = \Phi(S_{g,d,t})$, the three roles in group g have

$$Z_{g,r,d,t} = 2((U_{g,d,t} + r/3) \bmod 1) - 1, \quad r = 0, 1, 2.$$

The same phase shift applies to every coordinate. These population ranks have a continuous uniform distribution on the cube; they are not empirical ranks. Writing $u(z) = (z_1 + 1)/2$, define

$$\beta(z) = \sqrt{c_\beta} \begin{pmatrix} \sqrt{2/3} \cos(2\pi u(z)) \\ \sqrt{2/3} \sin(2\pi u(z)) \\ 1/\sqrt{3} \end{pmatrix}.$$

The trigonometric identities for three equally spaced phases imply $\|\beta(z)\|^2 = c_\beta$ and $B_t' B_t / N = \Gamma_\beta = (c_\beta/3)I_3$ at every date. Independent next-period shocks satisfy

$$F_{t+1} \sim N(\mu_F, I_3 - \mu_F \mu_F'), \quad \varepsilon_{t+1} \sim N(0, \sigma_\varepsilon^2 I_N), \quad R_{t+1} = B_t F_{t+1} + \varepsilon_{t+1}.$$

Thus $\mathbb{E}[F_{t+1}F'_{t+1}] = I_3$ is a second-moment normalization. The conditional mean and second moment of stock returns are $m_t = B_t\mu_F$ and $S_t = B_tB'_t + \sigma_\varepsilon^2 I_N$.

For the paper's weights $w_t(W) = W(Z_t)/N$, minimizing $\mathbb{E}[(1 - w_t(W)'R_{t+1})^2 \mid \mathcal{F}_t]$ gives $S_t w_t^\star = m_t$. The identity

$$(BB' + \sigma^2 I)^{-1}B = B(B'B + \sigma^2 I_3)^{-1}$$

and exact cross-sectional balance therefore yield

$$w_t^\star = \frac{1}{N}B_t \left(\Gamma_\beta + \frac{\sigma_\varepsilon^2}{N}I_3 \right)^{-1} \mu_F, \quad W^\star(z) = \frac{\beta(z)'\mu_F}{c_\beta/3 + \sigma_\varepsilon^2/N}.$$

The population policy is derived from the return model and the loss; it is not an independently imposed simulation target. With $g = c_\beta/3$,

$$q^\star = \frac{g}{g + \sigma_\varepsilon^2/N} \|\mu_F\|^2, \quad Q(W^\star) = 1 - q^\star, \quad (\text{SR}^\star)^2 = \frac{q^\star}{1 - q^\star}.$$

B. Assumption audit and spectrum

Stationary initialization and stable Gaussian AR dynamics give geometric beta-mixing; measurable rank and phase maps and independent next-period shocks preserve this property (E1). For $A_t = K_t/N^2$, the exact noncentral Gaussian quadratic-form MGF verifies

$$\log \mathbb{E} \left[\exp \left(R'_{t+1} A_t R_{t+1} / B_X^2 \right) \mid \mathcal{F}_t \right] \leq \frac{1}{2}, \quad B_X^2 = 4(c_\beta + \sigma_\varepsilon^2).$$

Conditional scalar Gaussian moments give $K_4 = 3$ (E2). The explicit sine, cosine, and constant loading functions have smooth extensions on the bounded cube. They belong to the Sobolev space of order $\nu + D/2 = 9/2$, equivalent to the Matérn-3/2 RKHS, as does their fixed linear combination $W^\star$ (E3). The conditional pricing restriction follows from $v'_t(m_t - S_t w_t^\star) = 0$ for every admissible v_t (E4).

For the uncentered managed second-moment operator,

$$\frac{\sigma_\varepsilon^2}{N} T_K \preceq \Sigma_{\mathcal{H}_K} \preceq \left(c_\beta + \frac{\sigma_\varepsilon^2}{N} \right) T_K.$$

The lower term is the idiosyncratic contribution; the upper bound follows by Jensen’s inequality and $\|\beta\|^2 = c_\beta$. Thus the managed spectrum has the Matérn exponent $b = 1 + 2\nu/D = 3/2$ (E5). The finite-sample audit uses one independent population quadrature with 3,072 nodes, including the within-triplet dependence, and tests the Loewner and ordered-eigenvalue bounds. Log-log fits use the fixed windows 30–200, 60–400, and 120–800. These finite-resolution fits do not prove an asymptotic exponent. The analytic optimal Sharpe is positive and finite and is compared with a 100,000-period stock-return simulation (E6).

C. Estimation, selection, and evaluation

We use the paper’s response-one ridge estimator in a controlled Nyström subspace of the actual Matérn RKHS, without rescaling the kernel diagonal. The inducing points are independent of all training and OOS samples. Empirical effective complexity is

$$\widehat{C}(\lambda) = \text{Tr}[\widehat{\Sigma}_T(\widehat{\Sigma}_T + \lambda I)^{-1}].$$

Training sample sizes and all 96 positive penalties are fixed before simulation. Every replication generates genuine stock-level returns. Sample-size paths are nested within a replication; different replications are independent. Validation uses the last quarter of the available history, following an earlier training block. After selecting lambda by validation loss, the policy is refitted on all available observations. An independent 720-period block gives OOS loss and unannualized Sharpe. Neither population moments, $W^\star$, nor OOS data enter selection. Separate independent population quadrature integrates conditional factor and idiosyncratic moments for more precise population-risk evaluation.

The population-risk oracle minimizes mean regret over the same fixed grid. Boundary choices and all high-complexity observations are retained. We report means, medians, pointwise Monte Carlo intervals, selection frequencies, and benchmarks for strong shrinkage, near-unregularized policies, a validated affine characteristic policy, and equal weights. Selected-penalty rate regressions use Monte Carlo means. The selection panel in the main complexity figure displays the median penalty; reporting both retains the effect of skewness and boundary choices rather than conflating the two.

For each penalty, the exact decomposition in the numerical subspace is

$$Q(\widehat{W}_\lambda) - Q(W^\star) = \{Q(W_\lambda) - Q(W^\star)\} + \|\widehat{W}_\lambda - W_\lambda\|_\Sigma^2 + 2\langle \widehat{W}_\lambda - W_\lambda, \Sigma W_\lambda - \mu \rangle.$$

The cross term is signed; only the squared-norm estimation term is necessarily nonnegative. Population bias includes the explicitly measured numerical approximation floor. Rank sensitivity compares 128, 256, and 512 inducing points. For numerical rank r , empirical complexity cannot exceed $\min(T, r)$. The near-unregularized end of each curve is therefore a subspace diagnostic and can be rank-sensitive. The paired acceptance check concerns oracle and validation-selected regret; it is not a uniform approximation guarantee over the entire penalty grid or an infinite-rank asymptotic experiment.

D. Monte Carlo evidence and limits

The headline uses 500 independent replications. At $T = 1440$, mean oracle and validation-selected population losses are 0.988906 and 0.991938; their mean population Sharpes are 0.10756 and 0.09595. Mean empirical complexities are 8.227 and 10.341. The selected-to-oracle mean regret ratio is 1.972. The validation boundary-selection frequency is 8.4 percent. The ensemble oracle selects a grid boundary at 0 of 10 sample sizes. These observations are retained in all plots and rate regressions. Along the full largest-T grid, mean independent OOS loss is 0.999977 at the lowest represented complexity and 1.198229 at the highest. Its observed grid minimum is 0.988305

at mean empirical complexity 8.227. The observed mean OOS Sharpe maximum is 0.11088 at complexity 8.227. These OOS extrema are descriptive and never determine a fitted policy or a validation choice. The oracle-regret log-log slope is -0.518 over the full grid (whole-replication bootstrap interval $[-0.535, -0.498]$) and -0.692 over its predeclared upper half ($[-0.742, -0.647]$). The difference between these windows shows the finite-sample transition; neither window nor the calibration was selected to match the theoretical slope.

Rate regressions use the full T grid, its upper half, and its largest four values. Bootstrap intervals resample entire independent replications, preserving their joint T-by-lambda surfaces and recomputing both the oracle and validation choices. The displayed theoretical slopes are benchmarks for the theory’s risk envelope; they are not imposed DGP-specific equalities for this fixed smooth population policy. Slower convergence, boundary selections, and differences from benchmark slopes are reported without modifying the calibration or fitting windows.

Robustness varies N independently of T, signal strength and idiosyncratic noise, characteristic persistence, and Matérn smoothness. All 15 additional configurations use 200 replications. Nearby multiples of three, $N \in \{99, 300, 501, 999\}$, preserve exact triplet balance. The weak/high-noise case uses $\mu_F/2$ and $\sigma_\varepsilon = 0.12$; the strong/low-noise case uses $2\mu_F$ and $\sigma_\varepsilon = 0.04$. Persistence is $\rho \in \{0.70, 0.90, 0.97\}$ and the alternative kernel smoothness values are $\nu \in \{0.5, 2.5\}$ at the same dimension $D = 6$. Empirical ranks are a separate finite-support experiment that violates the infinite-spectrum E5 condition. They use coordinatewise midranks $2(\text{rank}(Z_i) - 1/2)/N - 1$ and retain E1–E4 and E6. Bounded heteroskedastic Gaussian shocks retain tail control but generally destroy the baseline conditional pricing identity; their conditional standard deviation is $\sigma_\varepsilon(z) = 0.08 \exp(0.35z_2)$. E1 and E2 remain exact, and bounded positive volatility preserves the E5 operator comparison with adjusted constants. E3 and E6 for the unrestricted heteroskedastic optimum are not asserted. Their regret is explicitly measured against the numerical population optimum in the same Nyström subspace, not the baseline analytic optimum. The empirical JKP exercise is distinct from all of these artificial economies.

Table I. Baseline calibration and frozen Monte Carlo design.

Parameter	Value
N	600
T	60, 90, 120, 180, 240, 360, 540, 720, 1080, 1440
D	6
Economic factors	3
ρ	.95
c_β	0.0016
μ_F	(0.1, 0.06, 0.03)
σ_ε	0.08
Matérn ν	1.5
Length scale	1
Theoretical b	1.5
Population Sharpe	0.12008616
Nystrom rank	256
Headline replications	500
Independent OOS length	720

Table II. Baseline assumption audit. The spectral band is a finite-resolution diagnostic.

Condition	Theoretical status	Diagnostic	Tolerance	Result
E1	Exact: stationary stable Gaussian AR	spectral radius 0.950	< 1	Pass
E2(a)	Exact Gaussian quadratic MGF	max log MGF 0.0040657	<= 0.5	Pass
E2(b)	Conditional scalar Gaussian	max ratio 3	<= 3	Pass
E3	Smooth beta in Sobolev $H^{(9/2)}$	fixed finite linear combination	analytic	Pass
E4	Exact conditional normal equation	max residual 1.47e-17	< 1e-11	Pass
E5	Loewner sandwich; theoretical b=1.5	slope -1.2399	-1.5 +/- 0.35	Pass
E6	Analytic optimal Sharpe	MC 0.12297; exact 0.12009	abs error <= 0.0190	Pass

Table III. Monte Carlo mean population performance. Validation uses no population information; the oracle minimizes mean population regret over the fixed grid.

<i>A. Population performance</i>					
T	Oracle loss	Selected loss	Oracle SR	Selected SR	Regret ratio
60	0.999908	1.062600	0.01388	0.02365	5.439
90	0.999870	1.031790	0.01746	0.02912	3.266
120	0.999803	1.022915	0.02152	0.03762	2.649
180	0.999309	1.014299	0.06182	0.04641	2.108
240	0.997033	1.006801	0.07181	0.05002	1.868
360	0.993995	1.002833	0.08448	0.05938	2.076
540	0.991966	0.998236	0.09378	0.07131	2.014
720	0.990849	0.995804	0.09889	0.08095	1.978
1080	0.989672	0.993337	0.10413	0.09029	1.943
1440	0.988906	0.991938	0.10756	0.09595	1.972

<i>B. Effective complexity and regularization</i>				
T	Oracle C	Selected C	Oracle lambda	Selected lambda
60	0.1024	6.489	0.001129	0.003784
90	0.1597	6.008	0.0006952	0.003666
120	0.2433	6.355	0.0004281	0.003426
180	3.154	7.366	5.456e-06	0.003124
240	4.056	6.642	3.36e-06	0.00311
360	5.31	8.107	2.069e-06	0.002814
540	6.124	7.934	1.624e-06	0.002162
720	6.139	9.488	1.624e-06	0.001515
1080	7.101	10.13	1.274e-06	0.001016
1440	8.227	10.34	1e-06	0.0008655

Table IV. Fixed-window rate regressions with whole-replication bootstrap intervals. Theoretical slopes are benchmarks, not fitted targets.

Quantity	Window	Theory	Slope	OLS SE	CI low	CI high
Oracle lambda	Full	-0.6	-2.377	0.4304	-2.659	-2.311
Oracle lambda	Upper half	-0.6	-0.4902	0.06616	-0.5623	-0.4855
Oracle lambda	Largest four	-0.6	-0.5093	0.1171	-0.5093	-0.4221
Oracle C (pop.)	Full	0.4	1.441	0.2644	1.382	1.628
Oracle C (pop.)	Upper half	0.4	0.2903	0.04017	0.2875	0.3339
Oracle C (pop.)	Largest four	0.4	0.3046	0.07062	0.2529	0.3057
Oracle regret	Full	-0.6	-0.5177	0.05263	-0.5349	-0.4983
Oracle regret	Upper half	-0.6	-0.6915	0.0082	-0.7423	-0.6466
Oracle regret	Largest four	-0.6	-0.6903	0.01465	-0.7513	-0.6338
Oracle SR gap	Full	-0.6	-0.7272	0.03402	-0.7606	-0.7052
Oracle SR gap	Upper half	-0.6	-0.7466	0.01066	-0.8084	-0.6896
Oracle SR gap	Largest four	-0.6	-0.7485	0.01901	-0.8199	-0.6825
Selected lambda	Full	-0.6	-0.4691	0.06498	-0.5383	-0.4084
Selected lambda	Upper half	-0.6	-0.8976	0.05676	-1.107	-0.6996
Selected lambda	Largest four	-0.6	-0.9407	0.09274	-1.258	-0.6366
Selected C (pop.)	Full	0.4	0.06069	0.04431	0.01324	0.1133
Selected C (pop.)	Upper half	0.4	0.1829	0.05136	0.08263	0.2754
Selected C (pop.)	Largest four	0.4	0.2356	0.07656	0.09655	0.3714
Oracle C (sample)	Full	0.4	1.44	0.2629	1.379	1.625
Oracle C (sample)	Upper half	0.4	0.2956	0.03961	0.2925	0.3393
Oracle C (sample)	Largest four	0.4	0.3094	0.06971	0.2576	0.3108
Selected C (sample)	Full	0.4	0.1753	0.02188	0.1356	0.2191
Selected C (sample)	Upper half	0.4	0.2084	0.05085	0.1098	0.2989
Selected C (sample)	Largest four	0.4	0.2545	0.07946	0.1178	0.3886
Selected regret	Full	-0.6	-0.7604	0.02161	-0.7981	-0.7205
Selected regret	Upper half	-0.6	-0.7325	0.01028	-0.8027	-0.6673
Selected regret	Largest four	-0.6	-0.7154	0.007996	-0.8081	-0.6172
Selected SR gap	Full	-0.6	-0.4324	0.03721	-0.4555	-0.4098
Selected SR gap	Upper half	-0.6	-0.6729	0.0216	-0.7417	-0.6096
Selected SR gap	Largest four	-0.6	-0.7109	0.01212	-0.809	-0.6209

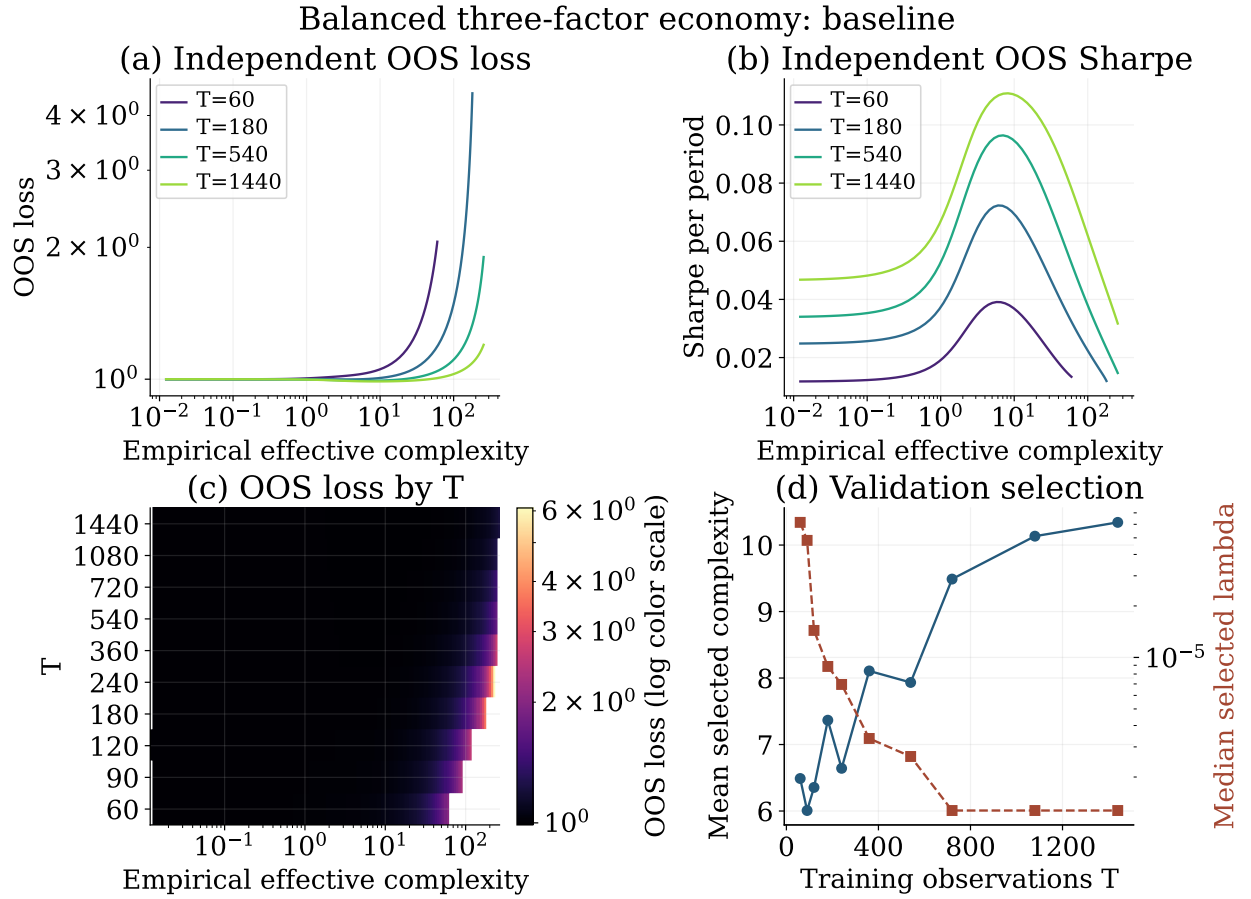


Figure 3. Independent OOS performance against empirical effective complexity. All penalties are retained; loss uses a logarithmic scale.

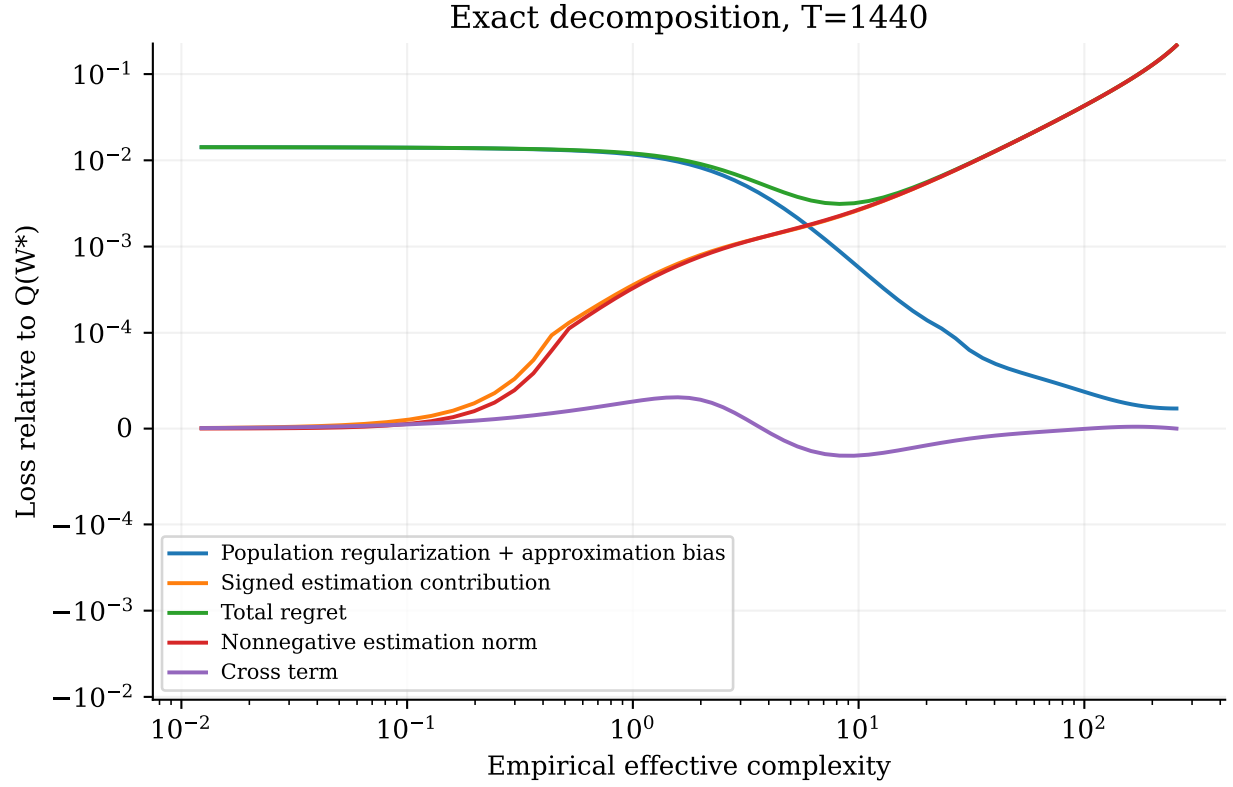


Figure 4. Exact bias, signed estimation contribution, and total regret. A symmetric logarithmic scale retains negative cross terms.

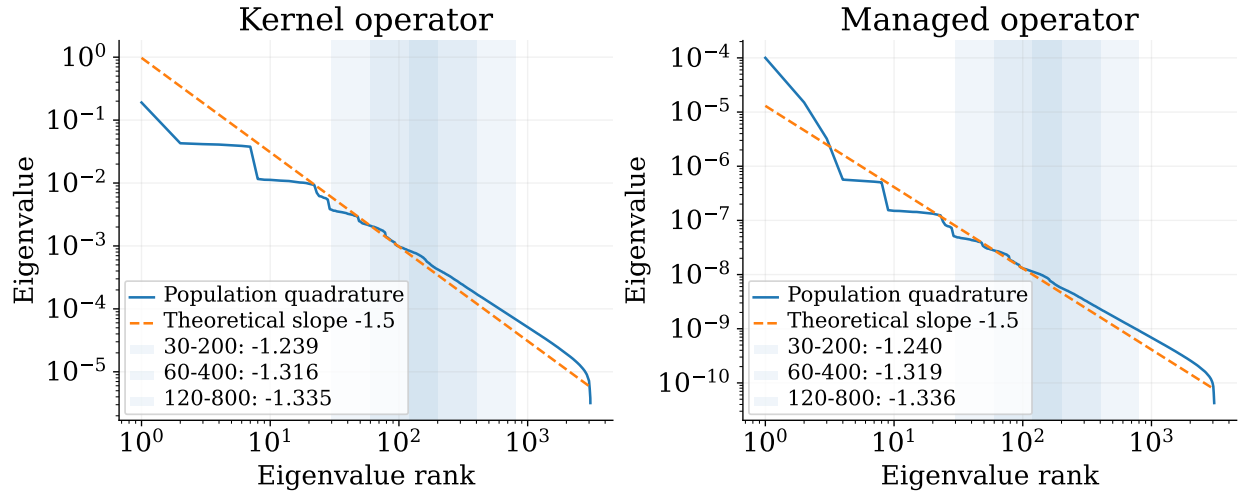


Figure 5. Population kernel and managed spectra on common independent quadrature. Shaded fixed windows show spectral-fit sensitivity.

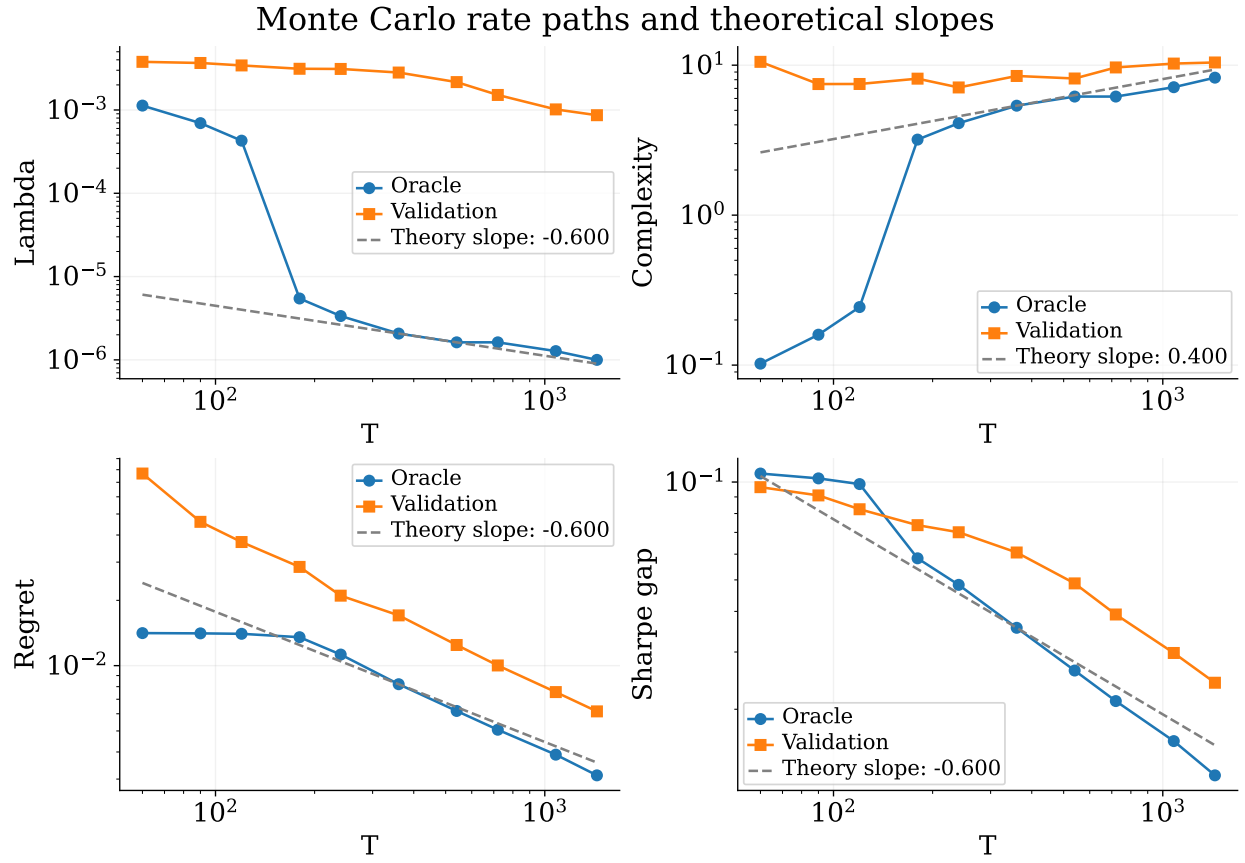


Figure 6. Oracle and validation rate paths. Dashed curves display theoretical slope benchmarks without constraining fitted results.

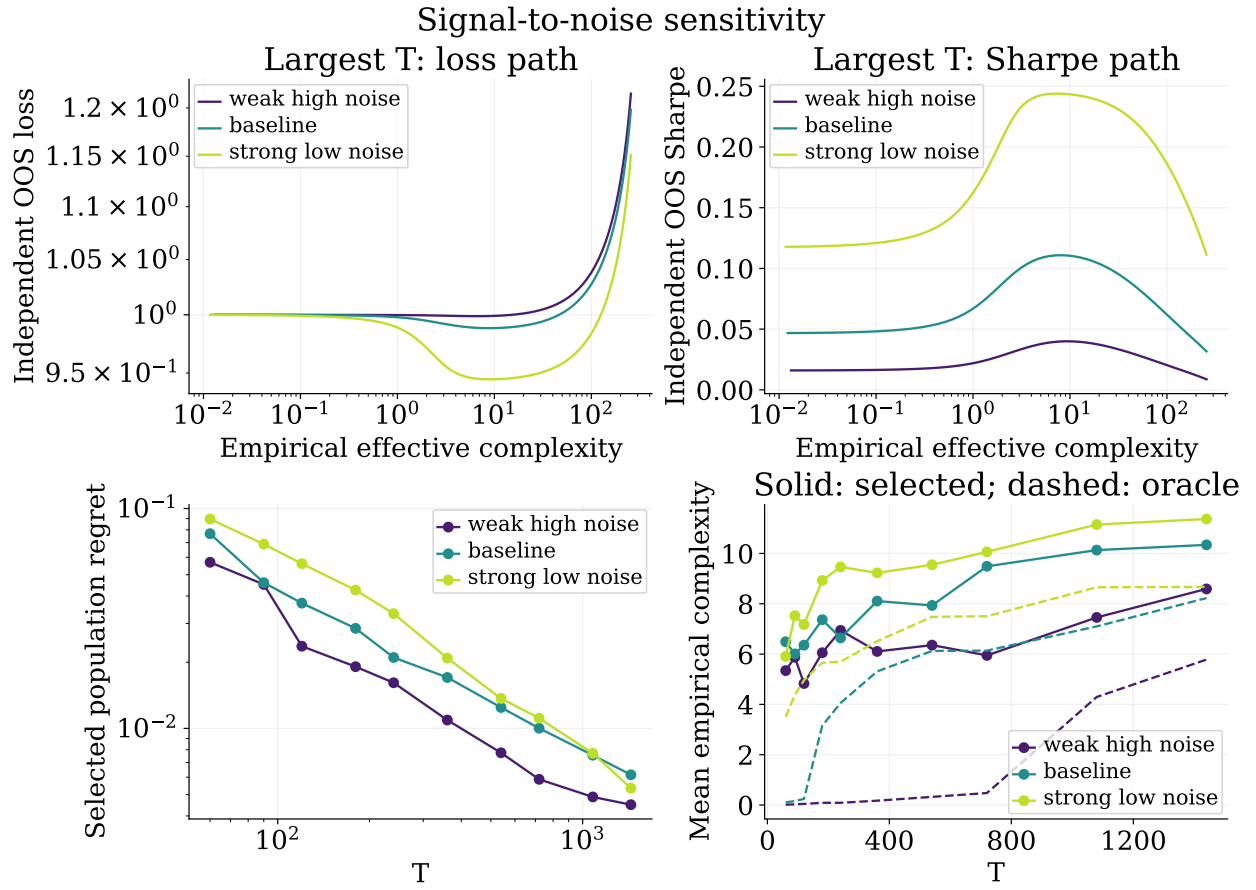


Figure 7. Signal-to-noise robustness with the same fixed learning design.

Table V. Robustness at the largest sample size. Heteroskedastic regret uses a finite-subspace reference; it is not a theorem-baseline result.

Environment	Oracle regret	Selected regret	Oracle C	Selected C	Selected SR	Boundary (%)
baseline	0.003121	0.006154	8.227	10.34	0.09595	8.4
N99	0.00563	0.008448	13.28	11.52	0.07596	17.5
N300	0.003778	0.006395	10.68	11.31	0.0933	4.5
N501	0.003095	0.005928	9.14	10.35	0.0952	7.5
N999	0.002623	0.005296	7.135	8.744	0.09924	6.5
Weak signal	0.002361	0.004499	5.781	8.59	0.02944	26.5
Strong signal	0.003241	0.005347	8.672	11.37	0.2388	0
$\rho = 0.70$	0.00315	0.006027	8.232	9.169	0.09655	7
$\rho = 0.90$	0.003072	0.005911	8.226	10.04	0.09701	7.5
$\rho = 0.97$	0.003169	0.006116	8.222	10.97	0.09664	6.5
$\nu = 0.5$	0.003274	0.006168	7.803	10.95	0.0963	5
$\nu = 2.5$	0.002692	0.005695	8.954	10.77	0.0997	6
rank128	0.003385	0.006611	8.319	9.921	0.09288	10
rank512	0.003187	0.006367	9.116	11.79	0.09407	10
Empirical ranks	0.003204	0.006074	8.198	10.09	0.09571	8
Heteroskedastic	0.003076	0.006411	8.513	10.31	0.09547	7

VII. Empirical analysis

We use the U.S. stock-level data of Jensen, Kelly, and Pedersen (2023). Inspired by Section 2.5 of Didisheim, Ke, Kelly, and Malamud (2024), we retain 130 of the 153 characteristics by coverage using only the initial training sample, 1963–1972. Validation and test observations cannot affect selection. The published candidate dictionary is a fixed retrospective research specification; we do not claim a historical data vintage or contemporaneous signal discovery. We retain common stocks on the main U.S. exchanges, excluding nano stocks, apply the 30% row-missingness threshold, and exclude stock-months with missing JKP next-month excess returns before monthly ranking and the calculation of N_t . This complete-case convention is our sample choice, not a rule attributed to Didisheim et al.; the sample is conditional on future payoff availability. We rank observed characteristics into $[-0.5, 0.5]$ and assign residual missing characteristics neutral zero. We do not impute missing payoffs.

Training expands from 1963, with the preceding five formation years reserved for validation and annual refits at the January formation close. The experiment contains 47 test years and 564 formation months, January 1978–December 2024; realized payoffs run from February 1978 to January 2025. Ridge penalties are selected using the portfolio criterion introduced earlier, along 120-point grids fixed using only the initial 1963–1972 managed-payoff spectra.

Linear is the finite-dimensional benchmark. Gaussian represents smooth nonlinear policies; Matérn-3/2 permits a broader nonlinear class. Each nonlinear policy uses one fixed map of 10,000 random Fourier features, seed zero. Both share a lengthscale of 4.3921, the median distance among 1,000 initial-training vectors. A common fixed positive scale targets median gross exposure 1.8 for the initial Linear validation policy. The last validation payoff becomes known at the January 1978 formation close, before the first OOS payoff.

Figure 8 and Table VI summarize the new run. Linear: Sharpe 3.09, maximum drawdown -15.8%; Gaussian: Sharpe 3.30, maximum drawdown -15.6%; Matérn-3/2: Sharpe 3.46, maximum drawdown -13.6%. Total wealth compounds the cash return plus the scaled portfolio excess payoff,

adding cash exactly once. Drawdown uses exactly that wealth path, including its starting value.

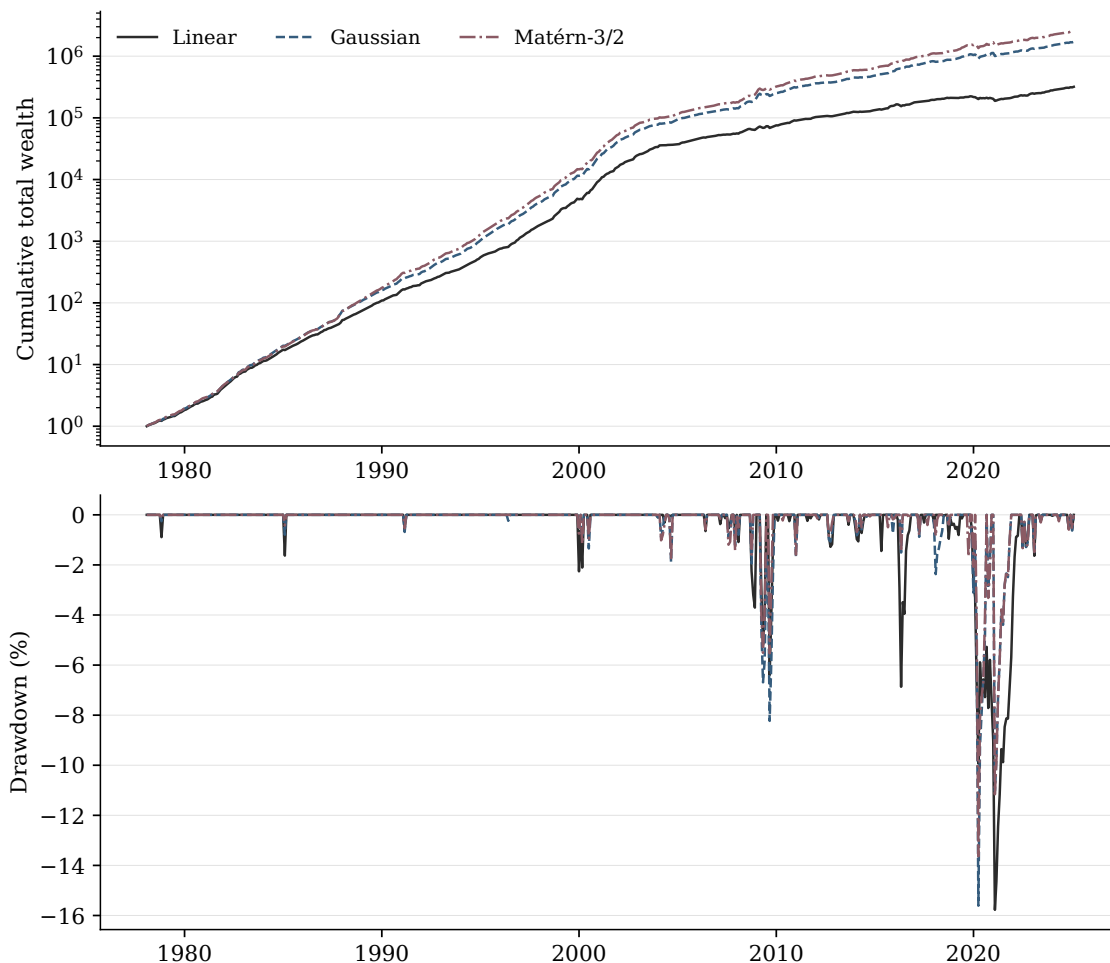


Figure 8. Cumulative total wealth (log scale) and drawdown from the same wealth series. All policies use the same fixed pre-payoff scale.

Figure 9 reports gross and net stock exposure under the same scale.

Representation determines which investment opportunities can potentially be expressed. The managed-portfolio spectrum describes their payoff structure. Effective complexity determines how much of that spectrum is actually used after regularization. Figure 10 shows the ordered positive second-moment eigenvalues on the common history preceding the 2024 test window.

Figures 11 and 12 plot historical and subsequent 2024 test Sharpe against $C(\lambda) = \sum_j \mu_j / (\mu_j + \lambda)$

Policy	Annual mean excess	Annual vol.	Sharpe	Max. drawdown	Mean gross	Mean net
Linear	0.234	0.076	3.091	-0.158	2.009	0.132
Gaussian	0.272	0.082	3.300	-0.156	2.326	0.100
Matérn-3/2	0.280	0.081	3.461	-0.136	2.374	0.090

Table VI. Performance from the 564 monthly OOS observations. Returns and volatility are annualized from decimal monthly excess returns; maximum drawdown is signed.

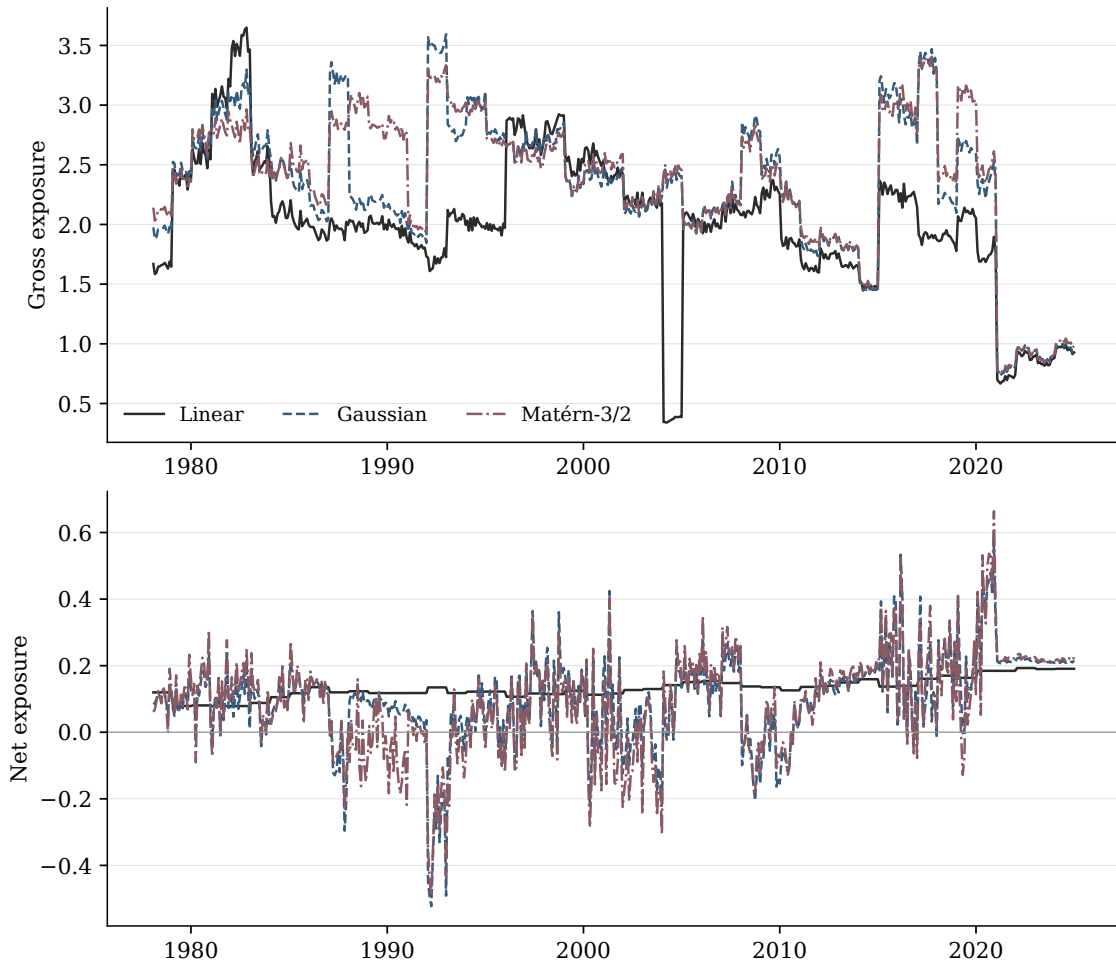


Figure 9. Gross and net exposures for the three policies.

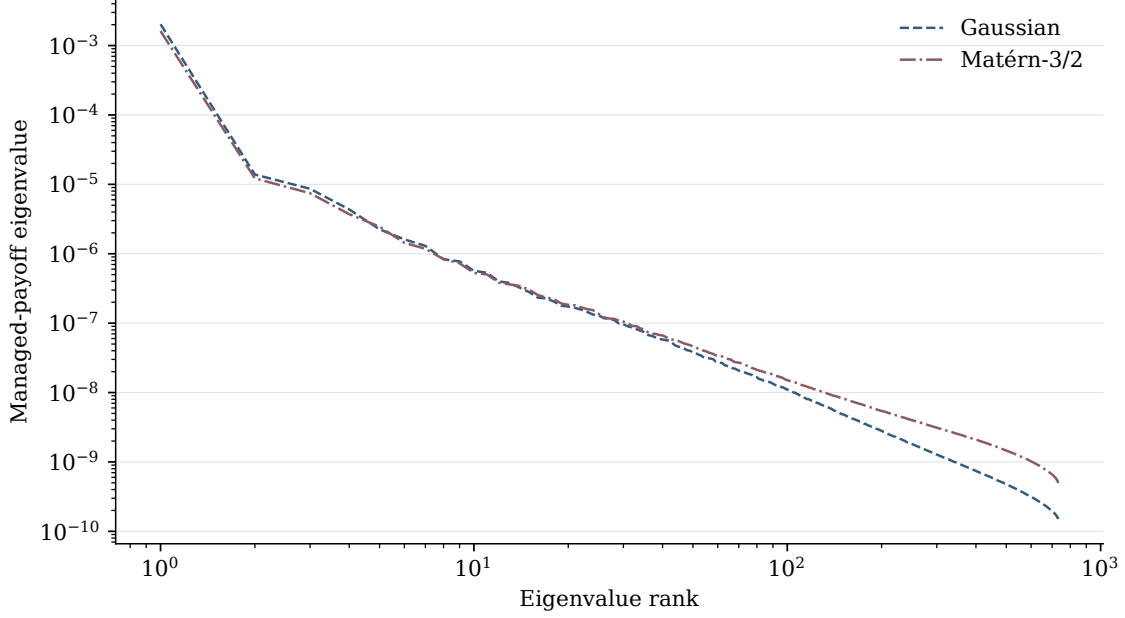


Figure 10. Managed-portfolio spectra on log-log axes.

along the same frozen penalty paths. The test panels use only the twelve payoffs from February 2024 through January 2025. Validation selections are distinguished from descriptive ex-post maxima, which never determine the policy.

An initially calibrated regularization schedule. For the Matérn-3/2 representation, we choose C_0 once using the 1963–1972 training and 1973–1977 formation-validation samples, and use

$$\lambda_T = C_0 T^{-\widehat{b}_T/(\widehat{b}_T+1)}.$$

Here T counts historical monthly managed payoffs, and $\widehat{b}_T$ is re-estimated at each January formation close together with the portfolio coefficients. It is the negative OLS slope of log second-moment eigenvalues on log rank, using the fixed 10%–60% rank band after a relative 10^{-12} numerical cutoff. This finite-sample estimation convention supplements the manuscript’s asymptotic rate; it does not establish its population spectral assumption. The main-text rate corresponds to $r = 1$

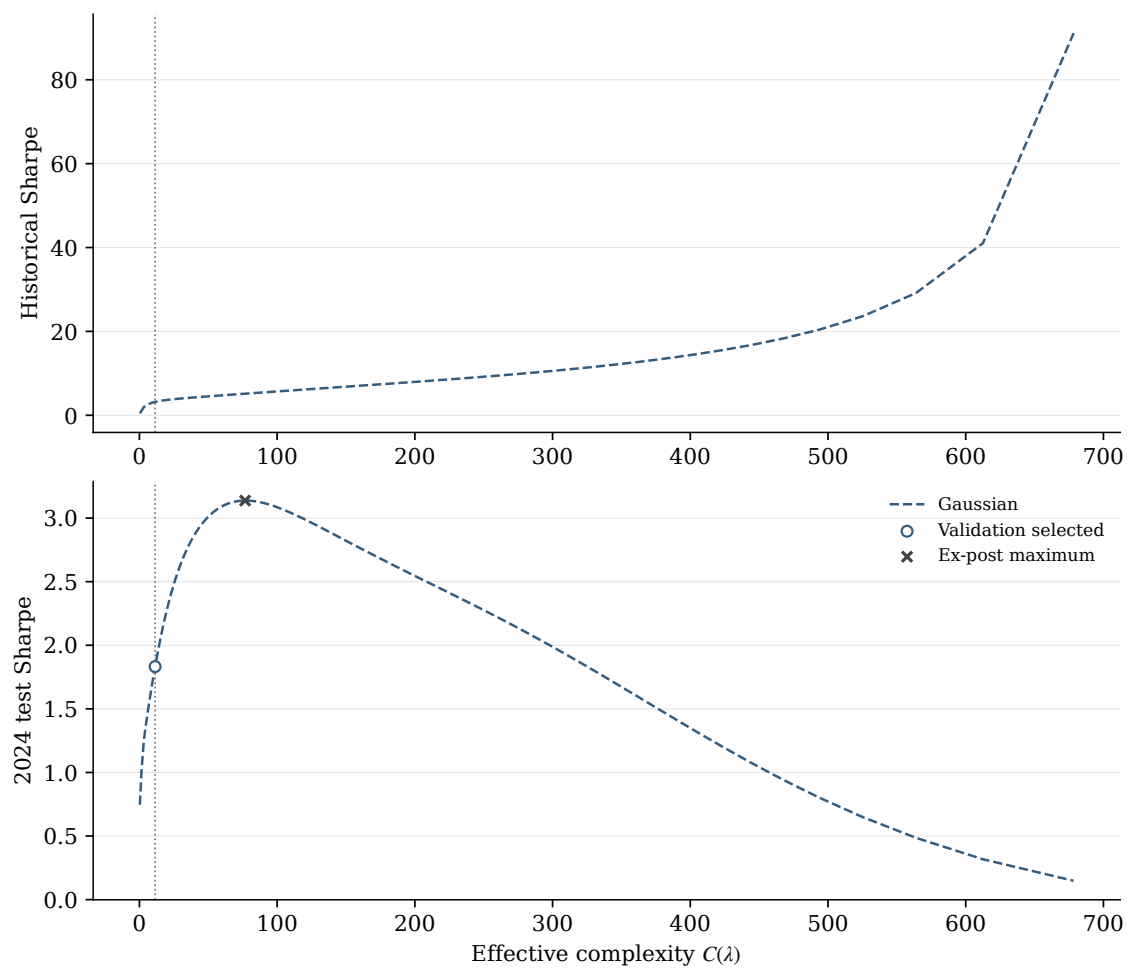


Figure 11. Gaussian effective complexity: historical and true 2024 test Sharpe.

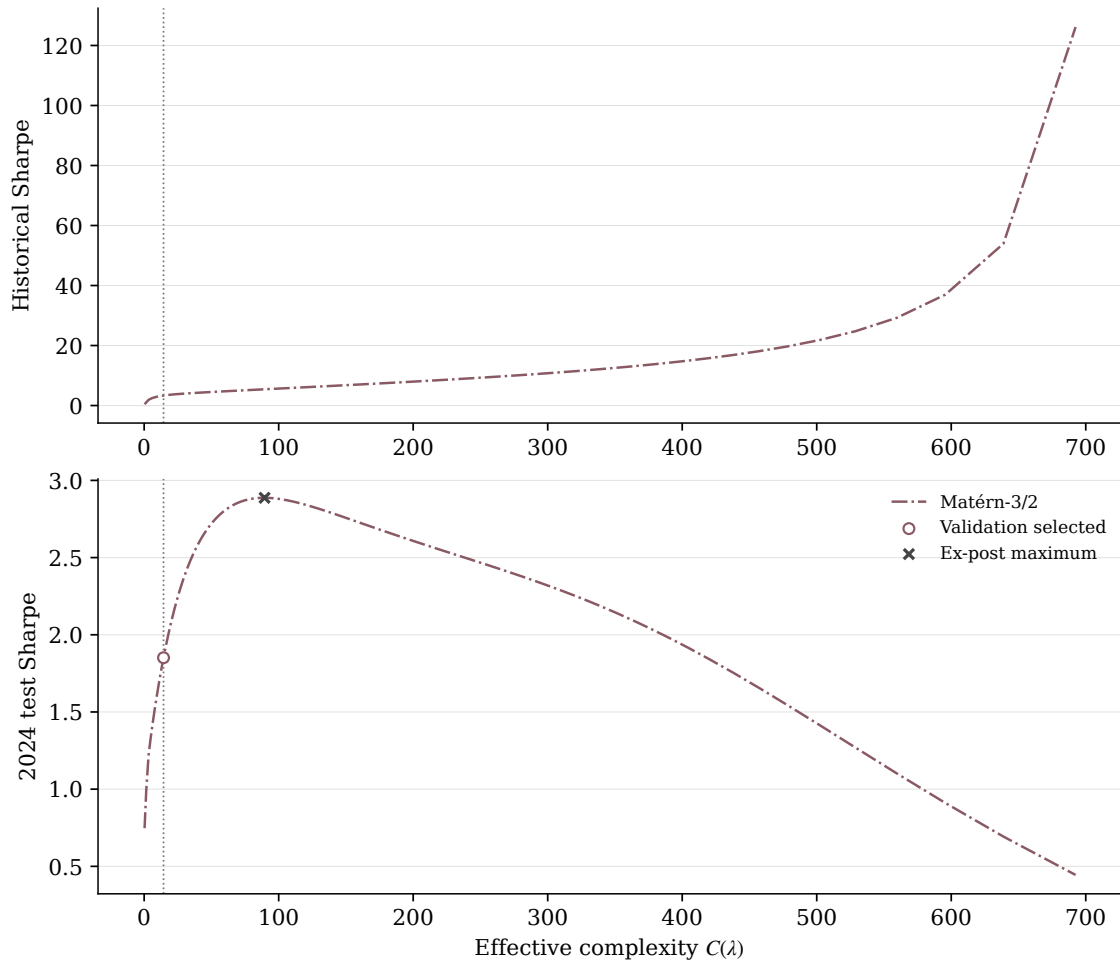


Figure 12. Matérn-3/2 effective complexity: historical and true 2024 test Sharpe.

in the general appendix. The grid retains all 120 previous initial-training penalties and adds 80 intermediate effective-complexity points before converting penalties to C_0 . Validation selects $C_0 = 2.265779 \times 10^{-6}$, available on January 31, 1978. The constant remains unchanged during all 47 subsequent annual refits.

Selected effective complexity rises from 36.681 in 1978 to 91.776 in 2024; regularization falls from 1.15864×10^{-7} to 4.67964×10^{-8} . Annual spectral estimates range from 1.3309 to 1.4323. The 564 selected OOS monthly payoffs have an annualized Sharpe of 3.561. This remains an exceptionally strong gross backtest, subject to the same complete-case payoff selection, retrospective characteristic dictionary, current-snapshot and execution assumptions as the main experiment; trading and borrowing costs are not subtracted.

Each annual curve evaluates only its twelve subsequent payoffs, from February through the following January. Annual Sharpe estimates are therefore noisy. Loss is the unscaled held-out mean of $(1 - R^p)^2$, with no test-period ridge penalty and no wealth-plot scaling. Neither the OOS loss nor OOS Sharpe selects C_0 , the spectral rank band, or the annual coefficients. Changes in estimated b could make the plug-in regularization path locally nonmonotonic, although no annual increases occur in this sample. No curve shape is imposed.

The displayed paths use 1,202 diagnostic values per year over an interval extended by a factor of 1,000 at each end, while the initial policy selection continues to use the frozen 200 candidates. A validation-only search over 3,001 points and numerical refinement improves the selected validation loss by only 0.0004464%. Additional evaluations at all 56,447 diagnostic midpoints bound the largest observed interpolation discrepancy at 0.00112 Sharpe points and 0.000057 loss units. The exact $\lambda = 0$ minimum-norm solution is included and marked on every curve; its empirical complexity is the historical rank, equal to T in all 47 windows. Its initial-validation loss is 0.820087, compared with 0.598552 at the selected positive penalty. Both weak- and strong-regularization limits are checked separately. Extrema at a boundary are not claimed to be interior optima. Companion figures divide complexity by each fit's historical monthly sample size T : the selected ratio falls from 0.2038 to 0.1254.

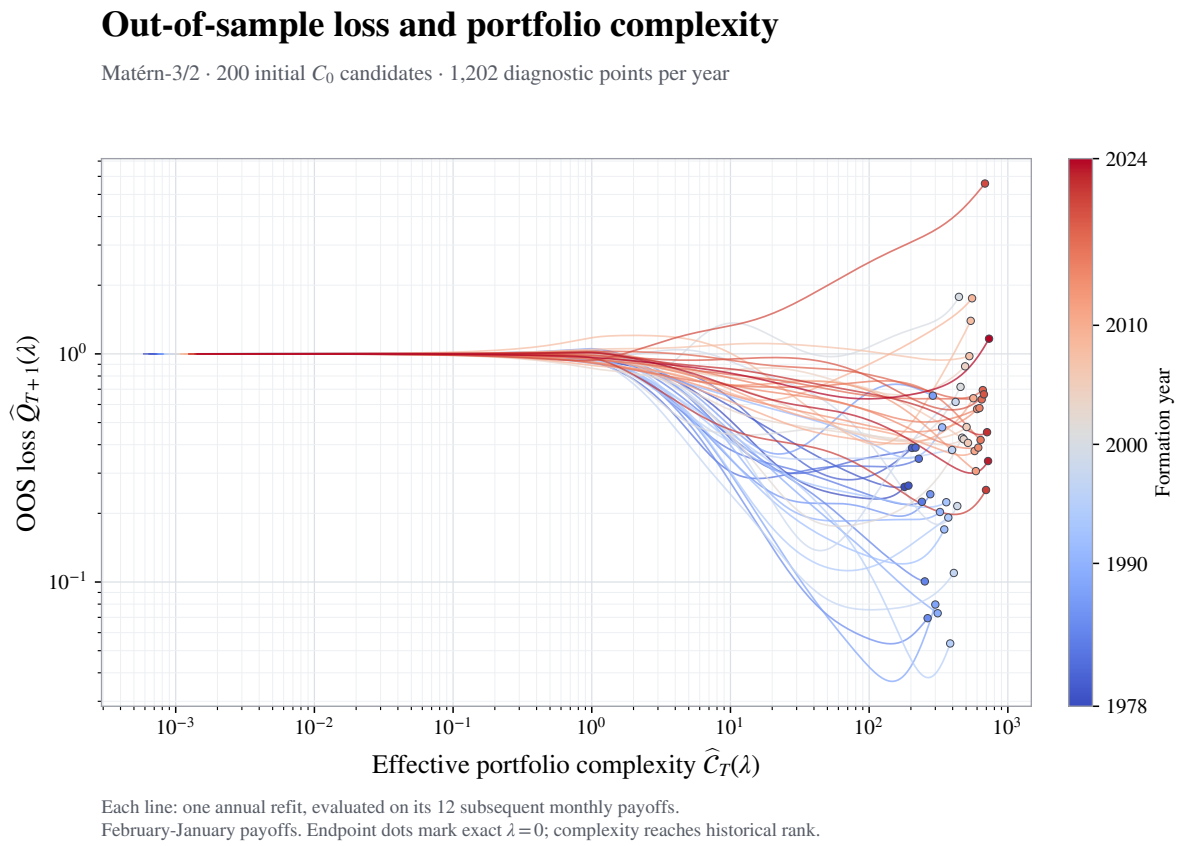


Figure 13. Next-year OOS quadratic loss against effective complexity for every annual Matérn refit, 1978–2024. Colors indicate formation years.

Out-of-sample Sharpe and portfolio complexity

Matérn-3/2 · 200 initial C_0 candidates · 1,202 diagnostic points per year

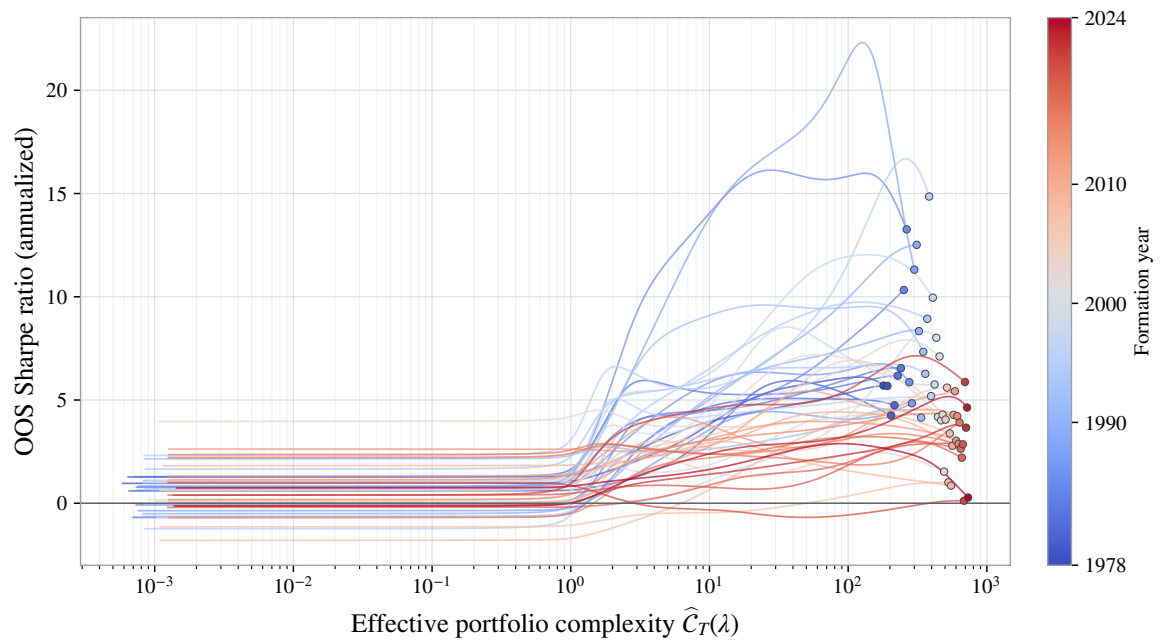
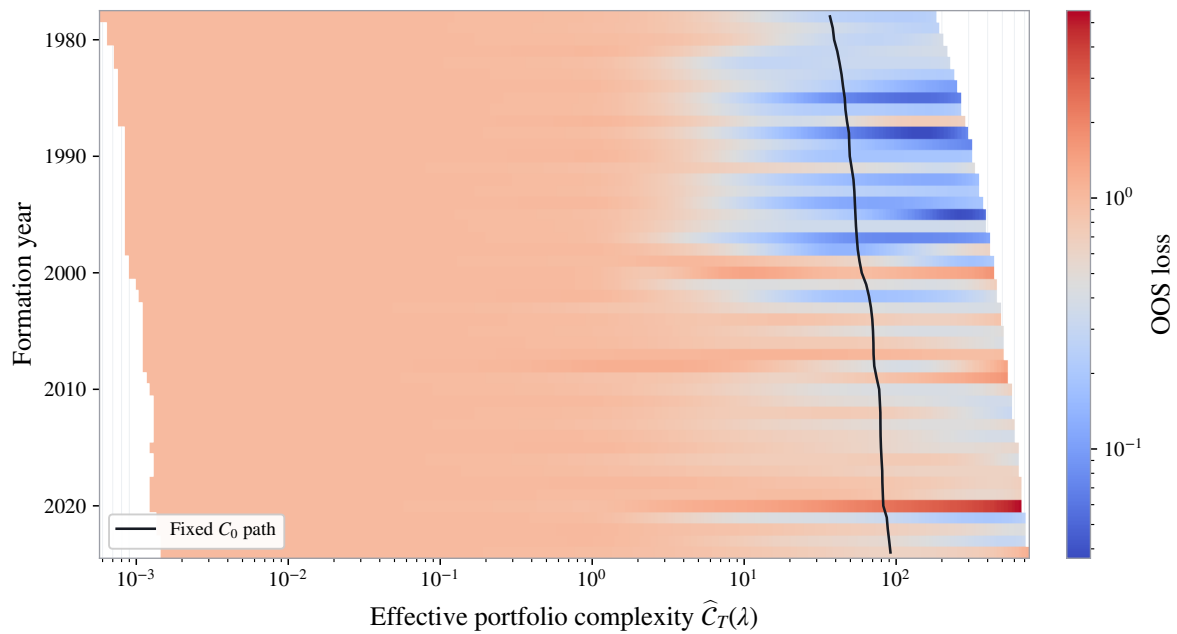


Figure 14. Annualized Sharpe against effective complexity, computed separately from each twelve-month OOS window.

The out-of-sample loss landscape

Matérn-3/2 · 200 initial C_0 candidates · 1,202 diagnostic points per year



Loss interpolated in log complexity within each year; blank cells lie outside observed support.
 Black path uses C_0 selected once in the initial validation; OOS losses never select it.

Figure 15. OOS loss by formation year and effective complexity. Interpolation is confined to each year's observed complexity support. The black line is the path implied by the initially selected, fixed C_0 .

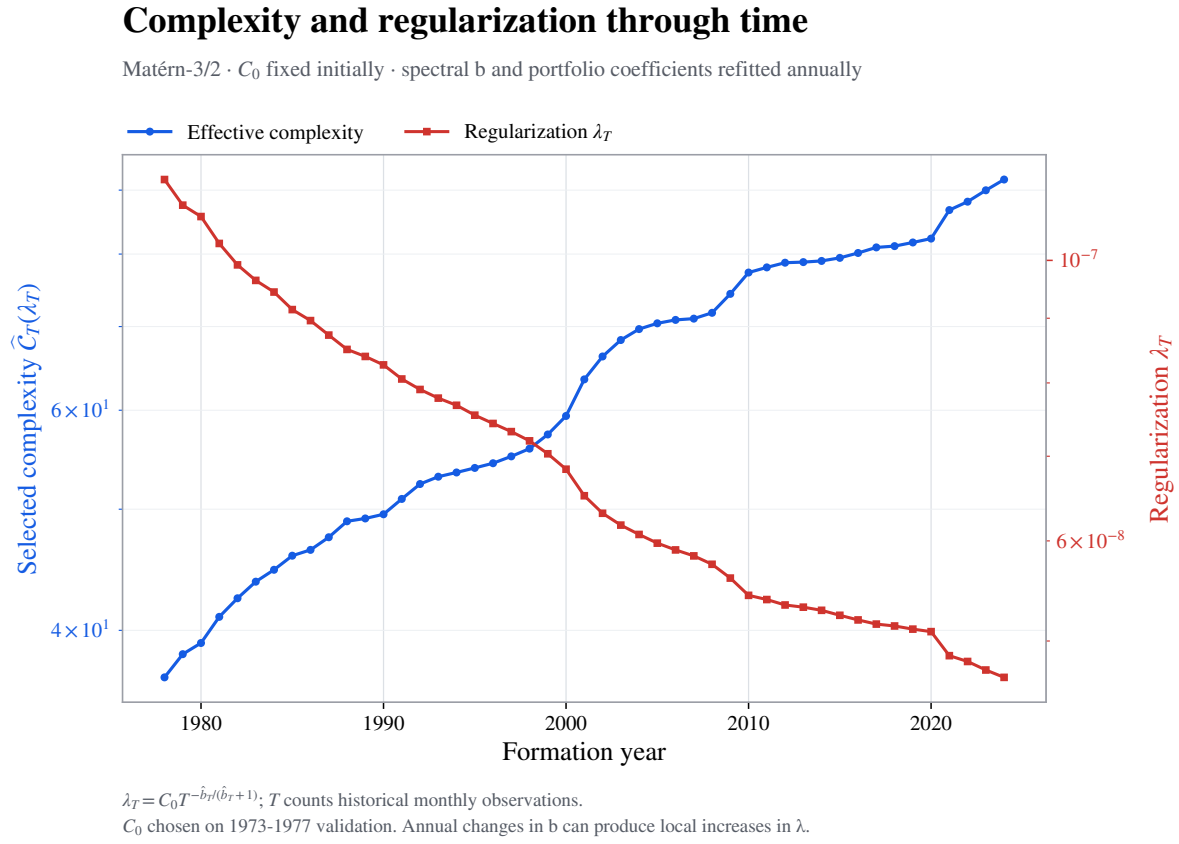


Figure 16. Effective complexity and regularization implied by a fixed initial C_0 and annually updated historical spectral estimates.

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A. Proofs and Auxiliary Results

A.1. Maximum-Sharpe direction

Proof of Proposition 1. Step 1: separate direction and scale. Every nonzero $W \in \mathcal{H}$ can be written as $W = aU$, with $\|U\|_{\mathcal{H}} = 1$ and $a \in \mathbb{R}$. Conversely, every such aU belongs to $\mathcal{H}$, and $a = 0$ includes the zero policy. Thus this parametrization does not change the feasible set. Since the payoff is linear in the policy,

$$\min_{W \in \mathcal{H}} Q(W) = \min_{\substack{U \in \mathcal{H} \\ \|U\|_{\mathcal{H}}=1}} \min_{a \in \mathbb{R}} \mathbb{E} \left[\left(1 - aR_{t+1}^p(U) \right)^2 \right].$$

A zero-payoff direction has loss one for every scale. A nonzero constant payoff would give minimum loss zero after rescaling. The assumption $0 < Q(W^*) < 1$ therefore allows zero-variance directions to be omitted.

Step 2: optimize the scale. Fix a direction U with positive payoff variance and write

$$m_U := \mathbb{E}[R_{t+1}^p(U)], \quad s_U^2 := \mathbb{E}[(R_{t+1}^p(U))^2] > 0.$$

Expanding the square and differentiating gives

$$Q(aU) = 1 - 2am_U + a^2s_U^2, \quad \frac{\partial Q(aU)}{\partial a} = -2m_U + 2as_U^2.$$

The second derivative is $2s_U^2 > 0$, so the unique minimizing scale is $a^*(U) = m_U/s_U^2$. Substitution

yields

$$\begin{aligned}
\min_{a \in \mathbb{R}} Q(aU) &= 1 - 2 \frac{m_U}{s_U^2} m_U + \left(\frac{m_U}{s_U^2} \right)^2 s_U^2 \\
&= 1 - \frac{m_U^2}{s_U^2} \\
&= \frac{\text{Var}(R_{t+1}^p(U))}{\text{Var}(R_{t+1}^p(U)) + m_U^2} \\
&= \frac{1}{1 + \text{SR}^2(R_{t+1}^p(U))}.
\end{aligned}$$

Step 3: compare directions. Consequently,

$$\min_{W \in \mathcal{H}} Q(W) = \min_{\substack{U \in \mathcal{H} \\ \|U\|_{\mathcal{H}}=1}} \frac{1}{1 + \text{SR}^2(R_{t+1}^p(U))}.$$

The right-hand side is minimized by a maximum-squared-Sharpe direction, because $x \mapsto (1+x)^{-1}$ is strictly decreasing on $[0, \infty)$. The corresponding optimally scaled payoff has mean $m_U^2/s_U^2 > 0$, since the minimum is strictly below one. It therefore attains the positive maximum Sharpe ratio, not its negative orientation. $\square$

A.2. Pricing-kernel interpretation and Hansen–Jagannathan equality

Proof of Proposition 2. Fix an economy and suppress its subscript in this proof.

Step 1: portfolio orthogonality. For any $V \in \mathcal{H}$, expansion of the quadratic criterion gives

$$\begin{aligned}
Q(W^\star + aV) &= Q(W^\star) - 2a\mathbb{E}[(1 - R_{t+1}^p(W^\star))R_{t+1}^p(V)] \\
&\quad + a^2\mathbb{E}[(R_{t+1}^p(V))^2].
\end{aligned}$$

Optimality at $a = 0$ implies $\mathbb{E}[m_{t+1}^\star R_{t+1}^p(V)] = 0$ for every V . In particular, choosing $V = W^\star$ gives

$$\mathbb{E}[R_{t+1}^p(W^\star)] = \mathbb{E}[(R_{t+1}^p(W^\star))^2].$$

Since $m^\star = 1 - R^p(W^\star)$, it follows that

$$\mathbb{E}[m^\star] = \mathbb{E}[(m^\star)^2] = Q(W^\star) =: q^\star \in (0, 1).$$

Proposition 1 then gives $q^\star = 1/(1 + (\text{SR}^\star)^2)$.

Step 2: normalize the pricing kernel. For $\tilde{m} = m^\star/q^\star$,

$$\mathbb{E}[\tilde{m}] = 1, \quad \mathbb{E}[\tilde{m}^2] = \frac{1}{q^\star}, \quad \text{Var}(\tilde{m}) = \frac{1 - q^\star}{q^\star} = (\text{SR}^\star)^2.$$

Step 3: verify the minimum-variance property. Let d be any square-integrable, possibly signed, random variable with $\mathbb{E}[d] = 1$ and $\mathbb{E}[dR^p(V)] = 0$ for every $V \in \mathcal{H}$. Then

$$\mathbb{E}[dm^\star] = \mathbb{E}[d] - \mathbb{E}[dR^p(W^\star)] = 1.$$

Hence $\mathbb{E}[d\tilde{m}] = 1/q^\star = \mathbb{E}[\tilde{m}^2]$, and therefore

$$\text{Var}(d) = \text{Var}(\tilde{m}) + \mathbb{E}[(d - \tilde{m})^2] \geq (\text{SR}^\star)^2.$$

Thus $\tilde{m}$ attains the Hansen–Jagannathan bound on the specified managed excess-return space, with no positivity restriction. None of these steps asserts the stronger conditional restriction in Condition (E4). □

A.3. Managed-portfolio Markowitz representation

Proof of Proposition 3. Under the integrability conditions verified in Lemma 1, the reproducing property gives $R_{t+1}^p(W) = \langle W, X_t \rangle_{\mathcal{H}_K}$. Therefore

$$Q(W) = 1 - 2\langle W, \mu_{\mathcal{H}_K} \rangle_{\mathcal{H}_K} + \langle W, \Sigma_{\mathcal{H}_K} W \rangle_{\mathcal{H}_K}.$$

For an arbitrary direction $V \in \mathcal{H}_K$,

$$DQ(W)[V] = 2\langle \Sigma_{\mathcal{H}_K} W - \mu_{\mathcal{H}_K}, V \rangle_{\mathcal{H}_K}.$$

The population first-order condition is consequently

$$\Sigma_{\mathcal{H}_K} W_K^\star = \mu_{\mathcal{H}_K}.$$

If the inverse is defined on $\mu_{\mathcal{H}_K}$ in the identifiable subspace, this gives the population representation in the proposition. The normal equation itself does not require a bounded inverse on all of $\mathcal{H}_K$.

The empirical penalized criterion is

$$1 - 2\langle W, \widehat{\mu}_T \rangle_{\mathcal{H}_K} + \langle W, \widehat{\Sigma}_T W \rangle_{\mathcal{H}_K} + \lambda \|W\|_{\mathcal{H}_K}^2.$$

Its derivative vanishes exactly when

$$(\widehat{\Sigma}_T + \lambda I) \widehat{W}_{\lambda, T} = \widehat{\mu}_T.$$

Since $\widehat{\Sigma}_T$ is positive and $\lambda > 0$, the operator on the left is boundedly invertible, with inverse norm at most $1/\lambda$. This yields the unique empirical policy in (5). $\square$

A.4. Consequences of the primitive conditions

Lemma 1 (Consequences of return-tail regularity). *Fix $\varepsilon \in \mathcal{E}$, and put $C_{\text{sc}} = \sqrt{8K_4(1 + K_4)}$. Write $\Sigma = \Sigma_{\mathcal{H}_K, \varepsilon}$.*

(i) *The feature $X_t = N^{-1} \sum_i R_{i,t+1} K(\cdot, Z_{i,t})$ is strongly measurable,*

$$R_{t+1}^p(W) = \langle W, X_t \rangle_{\mathcal{H}_K}, \quad \mathbb{E}_\varepsilon \left[\exp \left(\frac{\|X_t\|_{\mathcal{H}_K}^2}{B_X^2} \right) \middle| \mathcal{F}_t \right] \leq 2 \quad a.s.$$

In particular,

$$\mathbb{P}_\varepsilon(\|X_t\| > u \mid \mathcal{F}_t) \leq 2e^{-u^2/B_X^2} \quad a.s.,$$

and $\mathbb{E}_\varepsilon\|X_t\|^{2q} \leq 2q!B_X^{2q}$ for integers $q \geq 1$. These are moment bounds, not almost-sure bounds.

(ii) The operator Σ is positive, self-adjoint, trace class, and compact, with

$$\text{Tr } \Sigma = \mathbb{E}_\varepsilon\|X_t\|^2 \leq B_X^2 \log 2.$$

(iii) Writing $\mu = \mathbb{E}_\varepsilon X_t$, the minimum-norm target satisfies $\Sigma W_\varepsilon^\star = \mu$. For every $W \in \mathcal{H}_K$,

$$Q_\varepsilon(W) - Q_\varepsilon(W_\varepsilon^\star) = \|\Sigma^{1/2}(W - W_\varepsilon^\star)\|_{\mathcal{H}_K}^2.$$

(iv) Let $P_t^\star = R_{t+1}^p(W_\varepsilon^\star)$ and $q_t = \mathbb{E}_\varepsilon[(P_t^\star)^2 \mid \mathcal{F}_t]$. Condition (E4) implies

$$\mathbb{E}_\varepsilon[P_t^\star \mid \mathcal{F}_t] = q_t, \quad 0 \leq q_t \leq 1, \quad \mathbb{E}_\varepsilon[(m_{t+1,\varepsilon}^\star)^4 \mid \mathcal{F}_t] \leq 8(1 + K_4).$$

(v) The score $m_{t+1,\varepsilon}^\star X_t$ is a square-integrable Hilbert-valued martingale difference with respect to the economic timing:

$$\mathbb{E}_\varepsilon[m_{t+1,\varepsilon}^\star X_t \mid \mathcal{F}_t] = 0, \quad \mathbb{E}_\varepsilon[(m_{t+1,\varepsilon}^\star)^2 X_t \otimes X_t] \preceq C_{sc} \Sigma.$$

Proof. By the reproducing property,

$$\|X_t\|_{\mathcal{H}_K}^2 = \frac{1}{N^2} \sum_{i,j=1}^N R_{i,t+1} R_{j,t+1} K(Z_{i,t}, Z_{j,t}) = \frac{1}{N^2} R_{t+1}' K_t R_{t+1}.$$

Hence Condition (E2) gives part (i) directly. Markov's inequality yields the conditional tail bound,

and the nonnegative power series of the exponential yields the moment bound after taking expectations. Measurability follows from the measurable kernel sections and separability. Conditional Jensen followed by ordinary expectation gives $\mathbb{E}\|X_t\|^2 \leq B_X^2 \log 2$. Taking the expectation of the positive rank-one operator in trace norm proves part (ii).

Square integrability justifies differentiation of the population quadratic criterion. Its first-order condition is $\langle V, \mu - \Sigma W_\varepsilon^\star \rangle = 0$ for every $V \in \mathcal{H}_K$. Expanding around $W_\varepsilon^\star$ removes the cross term and proves part (iii). No bounded-support argument is used in these identities.

Taking $V = W_\varepsilon^\star$ in Condition (E4) yields $\mathbb{E}[P_t^\star \mid \mathcal{F}_t] = q_t$. Conditional Cauchy–Schwarz gives $q_t^2 \leq q_t$, hence $0 \leq q_t \leq 1$. The conditional fourth-moment bound in Condition (E2) applies to $P_t^\star$, since its stock weights are $\mathcal{F}_t$ -measurable. Thus

$$\mathbb{E}[(1 - P_t^\star)^4 \mid \mathcal{F}_t] \leq 8\{1 + \mathbb{E}[(P_t^\star)^4 \mid \mathcal{F}_t]\} \leq 8(1 + K_4 q_t^2) \leq 8(1 + K_4).$$

For every $V \in \mathcal{H}_K$, the same condition and conditional Cauchy–Schwarz give

$$\begin{aligned} \mathbb{E}[(m^\star)^2 (R^p(V))^2 \mid \mathcal{F}_t] &\leq \sqrt{\mathbb{E}[(m^\star)^4 \mid \mathcal{F}_t] \mathbb{E}[(R^p(V))^4 \mid \mathcal{F}_t]} \\ &\leq C_{\text{sc}} \mathbb{E}[(R^p(V))^2 \mid \mathcal{F}_t]. \end{aligned}$$

Integration proves the operator inequality in part (v). Summing over an orthonormal basis gives $\mathbb{E}\|m^\star X_t\|^2 \leq C_{\text{sc}} \text{Tr } \Sigma < \infty$. The conditional expectation therefore exists. Taking its inner products against a countable dense subset of $\mathcal{H}_K$ proves the martingale-difference identity from Condition (E4). $\square$

Proposition 4 (A sufficient Gaussian return specification). *Suppose that, conditionally on $\mathcal{F}_t$, R_{t+1} is Gaussian, possibly with a singular covariance, and*

$$\max_{1 \leq i \leq N} \mathbb{E}[R_{i,t+1}^2 \mid \mathcal{F}_t] \leq V_R < \infty \quad a.s.,$$

uniformly in the economy and time. Then Condition (E2) holds with $B_X^2 = 8\kappa_K V_R$ and $K_4 = 3$.

Proof. For every $V \in \mathcal{H}_K$, $R_{t+1}^p(V)$ is conditionally Gaussian. A scalar $G \sim N(m, v)$ satisfies

$$\mathbb{E}G^4 = m^4 + 6m^2v + 3v^2 \leq 3(m^2 + v)^2,$$

which proves (8) with $K_4 = 3$.

For the tail condition, Cauchy–Schwarz in $\mathcal{H}_K$ and $K(z, z) \leq \kappa_K$ give

$$\|X_t\|_{\mathcal{H}_K}^2 \leq \frac{\kappa_K}{N} \sum_{i=1}^N R_{i,t+1}^2.$$

If $G \sim N(m, v)$ and $m^2 + v \leq V_R$, then

$$\mathbb{E}e^{G^2/(8V_R)} = \left(1 - \frac{v}{4V_R}\right)^{-1/2} \exp\left(\frac{m^2}{8V_R - 2v}\right) \leq \frac{2}{\sqrt{3}}e^{1/6} < 2.$$

Therefore, by convexity,

$$\begin{aligned} \mathbb{E}_{\varepsilon} \left[\exp\left(\frac{\|X_t\|^2}{8\kappa_K V_R}\right) \middle| \mathcal{F}_t \right] &\leq \mathbb{E}_{\varepsilon} \left[\exp\left(\frac{1}{N} \sum_{i=1}^N \frac{R_{i,t+1}^2}{8V_R}\right) \middle| \mathcal{F}_t \right] \\ &\leq \frac{1}{N} \sum_{i=1}^N \mathbb{E}_{\varepsilon} \left[\exp\left(\frac{R_{i,t+1}^2}{8V_R}\right) \middle| \mathcal{F}_t \right] < 2. \end{aligned}$$

Independence across stocks is unnecessary. □

The proposition is a sufficient condition, not a claim that every Gaussian mixture has uniform covariance-relative sub-Gaussian tails. Conditional Gaussianity with bounded conditional second moments is enough here. No Bernstein condition is imposed after conditioning on the entire augmented input (Z_t, R_{t+1}) : at that point $m^\star$ is already known.

A.5. Exact reduction to the kernel learning problem

Lemma 2 (Induced KRR problem). *For every $\varepsilon \in \mathcal{E}$, the direct estimator is isometrically equivalent to scalar KRR with constant response one. Under this identification, the statistical covariance operator is $\Sigma_{\mathcal{H}_K, \varepsilon}$ and the KRR prediction error is exactly population portfolio regret. The source order is $r = 1$. Since the induced kernel need not be uniformly bounded, Theorem 3.2 of Bignozzi et al. (2026) is not invoked verbatim; Lemma 3 supplies the replacement empirical-stability and score bounds.*

Proof. Step 1: data and payoff representation. Let

$$\mathcal{X} := \mathcal{Z}^N \times \mathbb{R}^N, \quad \Xi_t := (Z_t, R_{t+1}).$$

For $x = ((z_1, \dots, z_N), r) \in \mathcal{X}$, put

$$\Phi(x) := \frac{1}{N} \sum_{i=1}^N r_i K(\cdot, z_i), \quad f_W(x) := \frac{1}{N} \sum_{i=1}^N W(z_i) r_i.$$

Then $\Phi(\Xi_t) = X_t$ and $f_W(\Xi_t) = \langle W, X_t \rangle_{\mathcal{H}_K} = R_{t+1}^p(W)$. The symbol Ξ_t denotes the augmented observation; X_t continues to denote the managed-payoff feature used in the main text.

Let $\mathcal{G} := \{f_W : W \in \mathcal{H}_K\}$ and equip it with $\langle f_W, f_V \rangle_{\mathcal{G}} := \langle W, V \rangle_{\mathcal{H}_K}$. This is well defined: if f_W vanishes on $\mathcal{X}$, choose $r = (N, 0, \dots, 0)$ and vary z_1 to obtain $W(z_1) = 0$. Thus the map $\mathcal{T} : W \mapsto f_W$ is a Hilbert-space isometry onto $\mathcal{G}$. This uses the declared input domain, not a full-support assumption on the actual return law.

With $Y_t \equiv 1$, the empirical objectives agree exactly:

$$\frac{1}{T} \sum_{t=1}^T |Y_t - f_W(\Xi_t)|^2 + \lambda \|f_W\|_{\mathcal{G}}^2 = \frac{1}{T} \sum_{t=1}^T (1 - R_{t+1}^p(W))^2 + \lambda \|W\|_{\mathcal{H}_K}^2.$$

This is an algebraic reformulation of the portfolio criterion, not an additional specification of the

return-generating process.

Step 2: induced kernel and population operator. The reproducing kernel of $\mathcal{G}$ is

$$\tilde{K}(x, x') = \langle \Phi(x), \Phi(x') \rangle_{\mathcal{H}_K} = \frac{1}{N^2} \sum_{i,j} r_i s_j K(z_i, z'_j), \quad x' = ((z'_1, \dots, z'_N), s).$$

It is measurable and $\tilde{K}(x, x) = \|\Phi(x)\|^2$, without a uniform bound on this diagonal. Nevertheless, $\mathbb{E}_\varepsilon \tilde{K}(\Xi_t, \Xi_t) < \infty$ by Lemma 1, so the canonical embedding is bounded. Let $J : \mathcal{G} \rightarrow L^2(\mathbb{P}_{\varepsilon, \Xi})$ be the canonical embedding. For $W, V \in \mathcal{H}_K$,

$$\langle J\mathcal{T}W, J\mathcal{T}V \rangle_{L^2} = \mathbb{E}_\varepsilon [\langle W, X_t \rangle \langle V, X_t \rangle] = \langle W, \Sigma_{\mathcal{H}_K, \varepsilon} V \rangle_{\mathcal{H}_K}.$$

Consequently,

$$\mathcal{T}^* J^* J \mathcal{T} = \Sigma_{\mathcal{H}_K, \varepsilon}, \quad \mathcal{T}^* J^* 1 = \mu_{\mathcal{H}_K, \varepsilon}.$$

The minimum-norm population target is $f_{\mathcal{G}} = \mathcal{T}W_\varepsilon^\star$, so $\|f_{\mathcal{G}}\|_{\mathcal{G}} \leq R$. This is precisely source order $r = 1$. The positive eigenvalues and effective dimension are unchanged by the isometry.

Step 3: dependence, filtration, and statistical residual. The space $\mathcal{X}$ is Polish, and $\{(\Xi_t, Y_t)\}$ inherits stationarity and the mixing bound from Condition (E1). Define its statistical history by

$$\mathcal{A}_t := \sigma(\Xi_s, Y_s : s \leq t), \quad \mathcal{B}_t := \mathcal{A}_{t-1} \vee \sigma(\Xi_t).$$

The economic timing in Definition 1 implies $\mathcal{A}_{t-1} \subseteq \mathcal{F}_t$. The statistical residual is

$$\xi_t := Y_t - f_{\mathcal{G}}(\Xi_t) = 1 - R_{t+1}^p(W_\varepsilon^\star) = m_{t+1, \varepsilon}^\star.$$

By Lemma 1(v) and the tower property,

$$\mathbb{E}_\varepsilon [X_t \xi_t \mid \mathcal{A}_{t-1}] = 0.$$

Since $\tilde{K}(\cdot, \Xi_t) = \mathcal{T}X_t$, the corresponding kernel-weighted residual in $\mathcal{G}$ is also a filtered innovation. Lemma 1 supplies the score variance bound. The residual ξ_t is a projection residual and is $\mathcal{B}_t$ -measurable. Consequently, an almost-sure conditional Bernstein bound given $\mathcal{B}_t$ would force it to be bounded; it is not imported from the companion theorem. We also do not assume $\mathbb{E}[\xi_t \mid \Xi_t] = 0$. The exact replacement for the probabilistic step is given next.

Step 4: identify the risk. For any fixed $W \in \mathcal{H}_K$, Lemma 1(iii) gives

$$\|Jf_W - Jf_{\mathcal{G}}\|_{L^2}^2 = \mathbb{E}_{\mathcal{E}}[(R_{t+1}^p(W) - R_{t+1}^p(W_{\mathcal{E}}^*))^2] = Q_{\mathcal{E}}(W) - Q_{\mathcal{E}}(W_{\mathcal{E}}^*).$$

The same identity applies to an estimated policy when the training data are held fixed in the population evaluation. All constants are inherited from the fixed representation and the parameters of $\mathcal{E}$. $\square$

A.6. Empirical stability with unbounded managed features

This subsection replaces the empirical-stability and noise proposition in Bignozzi et al. (2026). The covariance calculation uses their alternating-block/Berbee argument, with an analytical truncation of the feature norm. The score is controlled by a direct martingale second-moment argument. The estimator itself is not truncated, and the remaining operator decomposition and source-bias calculation are unchanged.

Lemma 3 (Unbounded-feature stability and score). *Assume Conditions (E1)–(E4) and $\text{Tr } \Sigma < \infty$.*

Put

$$C_{\text{sc}} := \sqrt{8K_4(1 + K_4)}.$$

For $0 < \lambda \leq 1$, put

$$D_{\lambda} = (\Sigma + \lambda I)^{-1/2}, \quad C_{\lambda} = D_{\lambda} \Sigma D_{\lambda}, \quad d_{\lambda} = \text{Tr } C_{\lambda} = C_K(\lambda),$$

$$\Delta_{\lambda,T} = D_{\lambda}(\Sigma - \widehat{\Sigma}_T)D_{\lambda}, \quad S_{\lambda,T} = \frac{1}{T} \sum_{t=1}^T D_{\lambda} m_{t+1}^{\star} X_t.$$

Fix $\delta \in (0, 1)$ and define

$$\begin{aligned} \ell &= \max\{1, \lceil (8C_{\beta}T/\delta)^{1/(\gamma+1)} \rceil\}, \\ H &= \log\left(\frac{eT}{\delta\lambda}\right), \quad \tau^2 = 8B_X^2 H, \quad B_{\lambda} = \frac{1 + \tau^2}{\lambda}, \\ z &= \log\left(\frac{112(1 + d_{\lambda})}{\delta}\right). \end{aligned}$$

If $\ell \leq T/4$, there is a numerical constant C such that, with probability at least $1 - \delta$,

$$\|\Delta_{\lambda,T}\|_{\text{op}} \leq C \left\{ \sqrt{\frac{\ell B_{\lambda} z}{T}} + \frac{\ell B_{\lambda} z}{T} \right\} + \frac{2B_X^2(8H + 1)}{\lambda} e^{-8H}, \quad (\text{A.1})$$

and

$$\|S_{\lambda,T}\|_{\mathcal{H}_K}^2 \leq \frac{2C_{\text{sc}}}{\delta} \frac{d_{\lambda}}{T}. \quad (\text{A.2})$$

Consequently, for each fixed δ there is a class-uniform C_{δ} such that, for sufficiently large T , the sufficient condition

$$T^{\gamma/(\gamma+1)} \lambda \geq C_{\delta} \left[\log\left(\frac{C_{\delta} T}{\lambda}\right) \right]^2 \quad (\text{A.3})$$

implies $\|\Delta_{\lambda,T}\|_{\text{op}} \leq 1/2$ and (A.2) with probability at least $1 - \delta$.

Proof. We suppress the economy index. Only the covariance is truncated in the proof. Let

$$A_t = (D_{\lambda} X_t) \otimes (D_{\lambda} X_t) \mathbf{1}_{\{\|X_t\| \leq \tau\}}, \quad C_{\lambda,\tau} = \mathbb{E} A_t, \quad H_t = A_t - C_{\lambda,\tau}.$$

The process (H_t) is centered and has no larger absolute-regularity coefficients than (Z_t, R_{t+1}) . Since $0 \preceq C_{\lambda,\tau} \preceq C_{\lambda} \preceq I$ and $\|A_t\|_{\text{op}} \leq \tau^2/\lambda$, we have

$$\|H_t\|_{\text{op}} \leq B_{\lambda}, \quad \mathbb{E} H_t^2 = \mathbb{E} A_t^2 - C_{\lambda,\tau}^2 \preceq B_{\lambda} C_{\lambda}. \quad (\text{A.4})$$

The second inequality is an operator inequality; no independence of successive observations is used.

Let $n = \lfloor T/(2\ell) \rfloor$ and split the first $2n\ell$ observations into $2n$ consecutive blocks of length ℓ . Treat the odd and even blocks separately. For an odd-block sum B_j^o , operator Cauchy–Schwarz gives

$$\|B_j^o\|_{\text{op}} \leq \ell B_\lambda, \quad \mathbb{E}(B_j^o)^2 \preceq \ell^2 B_\lambda C_\lambda.$$

Indeed, for every h , $\|\sum_{t \in I} H_t h\|^2 \leq \ell \sum_{t \in I} \|H_t h\|^2$. Berbee coupling replaces each family by independent blocks with identical block laws. The total probability of a coupling failure for both families is at most

$$2n\beta(\ell) \leq C_\beta T \ell^{-(\gamma+1)} \leq \delta/8.$$

For the independent copies, a variance majorant has operator norm at most $n\ell^2 B_\lambda$ and trace at most $n\ell^2 B_\lambda d_\lambda$. The trace form of the self-adjoint Bernstein inequality (Minsker, 2017, proof of Theorem 3.1 and Section 3.2), with this variance majorant, yields for each family, with failure probability at most $\delta/8$,

$$\left\| \frac{1}{T} \sum_{j=1}^n B_j^{o,*} \right\|_{\text{op}} \leq C \left\{ \sqrt{\frac{\ell B_\lambda z}{T}} + \frac{\ell B_\lambda z}{T} \right\}.$$

The trace-to-variance-majorant ratio is at most d_λ ; no bound on the effective rank of the actual variance is inferred by monotonicity. Apply the finite-dimensional inequality to increasing finite-rank projections and pass to the limit in operator norm, as in that reference. The operators here are trace class and bounded after truncation. The fewer than 2ℓ leftover observations contribute at most $2\ell B_\lambda/T$, already absorbed because $z \geq 1$.

It remains to compare the truncated and original covariances. Lemma 1 gives

$$\mathbb{P}\left(\max_{t \leq T} \|X_t\| > \tau\right) \leq 2T e^{-8H} \leq \delta/8.$$

On the complementary event, the truncated empirical covariance equals the original empirical

covariance. The population truncation bias satisfies

$$\begin{aligned}\|C_\lambda - C_{\lambda,\tau}\|_{\text{op}} &\leq \lambda^{-1} \mathbb{E}[\|X_t\|^2 \mathbf{1}_{\{\|X_t\| > \tau\}}] \\ &\leq \frac{2}{\lambda} (\tau^2 + B_X^2) e^{-\tau^2/B_X^2} = \frac{2B_X^2(8H+1)}{\lambda} e^{-8H}.\end{aligned}$$

The tail-integration formula justifies the second line. This proves (A.1) with total failure probability at most $\delta/2$.

For the score, Lemma 1 gives

$$\mathbb{E}\|D_\lambda m_{t+1}^\star X_t\|^2 \leq C_{\text{sc}} \text{Tr}(D_\lambda \Sigma D_\lambda) = C_{\text{sc}} d_\lambda.$$

For $s < t$, the score at s is $\mathcal{F}_t$ -measurable, whereas the conditional mean of the score at t given $\mathcal{F}_t$ is zero. All cross inner products consequently have zero expectation. Hence

$$\mathbb{E}\|S_{\lambda,T}\|^2 = \frac{1}{T^2} \sum_{t=1}^T \mathbb{E}\|D_\lambda m_{t+1}^\star X_t\|^2 \leq C_{\text{sc}} \frac{d_\lambda}{T}.$$

Markov's inequality proves (A.2) with failure probability $\delta/2$. There is no blocking loss in this score bound.

Finally, $d_\lambda \leq B_X^2 \log 2/\lambda$ and, for fixed δ , $\ell \leq C_\delta T^{1/(\gamma+1)}$. Thus the leading square-root term in (A.1) is bounded by a constant times

$$\left\{ \frac{[\log(C_\delta T/\lambda)]^2}{T^{\gamma/(\gamma+1)} \lambda} \right\}^{1/2}.$$

The linear term is bounded by the corresponding squared expression. The truncation bias tends to zero uniformly for $0 < \lambda \leq 1$ as $T \rightarrow \infty$. Increasing C_δ proves the final statement. $\square$

Corollary 1 (Finite-sample portfolio loss). *Under the assumptions of Lemma 3, for each fixed*

$\delta \in (0, 1)$ and sufficiently large T satisfying (A.3),

$$Q_\varepsilon(\widehat{W}_{\lambda,T}) - Q_\varepsilon(W_\varepsilon^\star) \leq 9R^2\lambda + C_\delta \frac{C_K(\lambda)}{T}$$

with probability at least $1 - \delta$, uniformly over the class.

Proof. Use the error decomposition, source-bias bound and operator-interaction bound of Bignozzi et al. (2026), specialized to $r = 1$. For completeness, write

$$\begin{aligned} W_\lambda &= (\Sigma + \lambda I)^{-1} \Sigma W^\star, & B &= W_\lambda - W^\star, \\ V_1 &= (\widehat{\Sigma}_T + \lambda I)^{-1} \frac{1}{T} \sum_t m_{t+1}^\star X_t, \\ V_2 &= (\widehat{\Sigma}_T + \lambda I)^{-1} (\Sigma - \widehat{\Sigma}_T) B. \end{aligned}$$

Then $\widehat{W} - W^\star = B + V_1 + V_2$. Spectral calculus gives $\|\Sigma^{1/2} B\|^2 \leq R^2 \lambda$ and $\|(\Sigma + \lambda I)^{1/2} B\|^2 \leq 2R^2 \lambda$. On the stability event,

$$\|\Sigma^{1/2} V_1\|^2 \leq 4\|S_{\lambda,T}\|^2, \quad \|\Sigma^{1/2} V_2\|^2 \leq 2R^2 \lambda.$$

The first inequality follows from $\|(I - \Delta_{\lambda,T})^{-1}\| \leq 2$; the second follows from $\|(I - \Delta_{\lambda,T})^{-1} \Delta_{\lambda,T}\| \leq 1$. Finally, $\|u + v + w\|^2 \leq 3(\|u\|^2 + \|v\|^2 + \|w\|^2)$ and Lemma 3 give the claim. In particular, one may take the score constant $24C_{\text{sc}}/\delta$. $\square$

A.7. From portfolio loss to the Sharpe ratio

Lemma 4 (Sharpe-ratio recovery without loss of exponent). *Let $\widehat{W}_T \in \mathcal{H}_K$ be any estimated policy satisfying, uniformly over $\varepsilon \in \mathcal{E}$,*

$$D_T := Q_\varepsilon(\widehat{W}_T) - Q_\varepsilon(W_\varepsilon^\star) = O_{\mathbb{P}_\varepsilon}(a_T), \quad a_T \longrightarrow 0.$$

Then the estimated payoff has positive mean and positive variance with probability tending to one, uniformly, and

$$\text{SR}_{\varepsilon,K}^* - \text{SR}(R_{t+1}^p(\widehat{W}_T)) = O_{\mathbb{P}_\varepsilon}(a_T).$$

Sharpe may be assigned any finite convention on a zero-variance event, whose probability tends to zero. No sign adjustment using unknown population moments is required.

Proof. Write $q^* = Q_\varepsilon(W_\varepsilon^*)$ and $S^* = \text{SR}_{\varepsilon,K}^*$. By Proposition 2 and Condition (E6),

$$\underline{q} := \frac{1}{1 + \underline{S}^2} \leq q^* \leq \frac{1}{1 + \underline{S}^2} =: \bar{q} < 1.$$

Step 1: recover the sign and exclude degeneracy. On the event $D_T < 1 - \bar{q}$, which has probability tending to one uniformly, $Q_\varepsilon(\widehat{W}_T) < 1$. Expanding this loss yields

$$2\mathbb{E}_\varepsilon[R_{t+1}^p(\widehat{W}_T)] > \mathbb{E}_\varepsilon[(R_{t+1}^p(\widehat{W}_T))^2] \geq 0.$$

Thus its mean is strictly positive. It cannot have zero variance: a zero-variance payoff with positive mean is a positive constant almost surely, and scaling the same policy would produce the unit payoff, contradicting $q^* \geq \underline{q} > 0$. Hence its Sharpe ratio $\widehat{S}_T$ is positive on this event.

Step 2: bound squared-Sharpe loss. Optimizing only the scale of $\widehat{W}_T$ gives

$$q^* \leq \frac{1}{1 + \widehat{S}_T^2} = \min_{a \in \mathbb{R}} Q_\varepsilon(a\widehat{W}_T) \leq Q_\varepsilon(\widehat{W}_T) = q^* + D_T.$$

Therefore

$$0 \leq (S^*)^2 - \widehat{S}_T^2 \leq \frac{1}{q^*} - \frac{1}{q^* + D_T} = \frac{D_T}{q^*(q^* + D_T)} \leq \frac{D_T}{\underline{q}^2}.$$

Step 3: pass to the level Sharpe. Since $S^\star + \widehat{S}_T \geq \underline{S}$ on the same event,

$$0 \leq S^\star - \widehat{S}_T = \frac{(S^\star)^2 - \widehat{S}_T^2}{S^\star + \widehat{S}_T} \leq \frac{D_T}{\underline{S} q^2}.$$

This proves the stated rate uniformly over $\mathcal{E}$. The complementary event has uniformly vanishing probability. $\square$

A.8. Proof of the portfolio-learning rate

Proof of Theorem 1. Fix $\varepsilon \in \mathcal{E}$.

Step 1: use the unbounded-feature extension. The proof follows the operator decomposition of Bignozzi et al. (2026), with its empirical-stability and noise step replaced by Lemma 3. Corollary 1 therefore gives, for every fixed $\delta \in (0, 1)$, a constant C_δ common to the class such that

$$Q_\varepsilon(\widehat{W}_{\lambda,T}) - Q_\varepsilon(W_\varepsilon^\star) \leq C_\delta \left(\lambda + \frac{C_{K,\varepsilon}(\lambda)}{T} \right)$$

with probability at least $1 - \delta$, for sufficiently large T satisfying

$$T^{\gamma/(\gamma+1)} \lambda \geq C_\delta [\log(C_\delta T / \lambda)]^2. \quad (\text{A.5})$$

The extra logarithm occurs only in this sufficient stability condition, not in the leading score variance. By Condition (E5) and Proposition A.6 of Bignozzi et al. (2026), $C_{K,\varepsilon}(\lambda) \asymp \lambda^{-1/b}$ uniformly.

Step 2: verify the admissible regularization scale. For $\lambda_T^\star \asymp T^{-b/(b+1)}$, the class restriction $\gamma > b$ implies

$$\frac{\gamma}{\gamma+1} - \frac{b}{b+1} = \frac{\gamma-b}{(\gamma+1)(b+1)} > 0.$$

Consequently,

$$\frac{T^{\gamma/(\gamma+1)} \lambda_T^\star}{[\log(C_\delta T / \lambda_T^\star)]^2} \longrightarrow \infty,$$

so (A.5) holds eventually. At this scale,

$$\lambda_T^\star \asymp \frac{C_{K,\varepsilon}(\lambda_T^\star)}{T} \asymp T^{-b/(b+1)}.$$

This proves the population-loss rate. Uniformity means that, for each $\delta \in (0, 1)$, some $C_\delta, T_0(\delta)$ independent of ε satisfy

$$\sup_{\varepsilon \in \mathcal{E}} \mathbb{P}_\varepsilon \left(Q_\varepsilon(\widehat{W}_{\lambda_T^\star, T}) - Q_\varepsilon(W_\varepsilon^\star) > C_\delta T^{-b/(b+1)} \right) \leq \delta, \quad T \geq T_0(\delta).$$

Step 3: translate loss into Sharpe and frontier recovery. Apply Lemma 4 with $a_T = T^{-b/(b+1)}$. At any fixed volatility target σ , the difference between the population and learned capital-allocation lines is

$$\mu_K^\star(\sigma) - \widehat{\mu}_{K,T}(\sigma) = \sigma \left(\text{SR}_{\varepsilon,K}^\star - \text{SR}(R_{t+1}^p(\widehat{W}_{\lambda_T^\star, T})) \right),$$

which has the same order, also uniformly for σ in a bounded interval. This concerns the tangency capital-allocation line with a risk-free asset and scalar rescaling, not the entire risky-only frontier. $\square$

A.9. Effective portfolio complexity

Proof of Theorem 2. The proof of Theorem 1 already establishes admissibility of $\lambda_T^\star$ and the spectral relation $C_{K,\varepsilon}(\lambda) \asymp \lambda^{-1/b}$. Thus

$$C_{K,T}^\star = C_{K,\varepsilon}(\lambda_T^\star) \asymp T^{1/(b+1)}, \quad \frac{C_{K,T}^\star}{T} \asymp T^{-b/(b+1)} \longrightarrow 0.$$

It is finite for every T , since $C_{K,\varepsilon}(\lambda) \leq \text{Tr}(\Sigma_{\mathcal{H}_{K,\varepsilon}})/\lambda$.

To express the same balance in complexity units, let $C = C_{K,\varepsilon}(\lambda)$, so that $\lambda \asymp C^{-b}$. The leading risk upper envelope has the form

$$\mathfrak{R}_T(C) = aC^{-b} + d\frac{C}{T}, \quad a, d > 0.$$

Its derivatives are

$$\mathfrak{R}'_T(C) = -baC^{-b-1} + \frac{d}{T}, \quad \mathfrak{R}''_T(C) = b(b+1)aC^{-b-2} > 0.$$

Hence its unique minimum occurs at

$$C_T^{\text{env}} = \left(\frac{ba}{d} T \right)^{1/(b+1)}.$$

This is an upper-envelope calculation, not a shape theorem for the exact Sharpe curve of every economy. No matching pointwise bias lower bound is used. Condition (E6) is not needed for this conclusion. $\square$

A.10. A finance-specific minimax construction

Proof of Theorem 3. The construction keeps the policy kernel fixed. It is a least-favourable family used only in this lower-bound proof, not a structural return equation imposed on the economies in the main text.

Step 1: construct a least-favourable family with nondegenerate Sharpe. Since $\mathcal{E}$ is nonempty, take the one-period state-return marginal ν_0 of one economy in the class. Write a draw as $x = (\mathbf{z}, r)$ and put

$$Q_X(x) := \|\Phi(x)\|_{\mathcal{H}_K}^2, \quad q_X := \int Q_X(x) d\nu_0(x).$$

Condition (E2) and Jensen's inequality imply $q_X \leq B_X^2 \log 2$. Moreover $q_X > 0$, because otherwise the managed feature would vanish almost surely, contradicting the spectral lower bound in Condition (E5).

Define a tilted law by

$$d\tilde{\nu}(x) = \frac{Q_X(x)}{q_X} d\nu_0(x),$$

and, on the support of the tilted law, replace the return vector by

$$r^\circ = \sqrt{\frac{q_X}{Q_X(x)}} r.$$

Let ν denote the law of $(\mathbf{z}, r^\circ)$ and reuse r for the normalized return vector below. Since the feature map is linear in returns,

$$\int \Phi(\mathbf{z}, r^\circ) \otimes \Phi(\mathbf{z}, r^\circ) d\tilde{\nu} = \int \Phi(x) \otimes \Phi(x) d\nu_0(x) =: \Sigma_0.$$

Thus the normalization preserves the entire managed-payoff second-moment operator. Under ν ,

$$\|\Phi(x)\|_{\mathcal{H}_K}^2 = q_X \leq B_X^2 \log 2 \quad \text{almost surely.}$$

Put $\bar{B}_X := B_X \sqrt{\log 2}$.

Write

$$\Sigma_0 e_j = \mu_j e_j, \quad c_\mu j^{-b} \leq \mu_j \leq C_\mu j^{-b}.$$

Its eigenvectors are orthonormal in $\mathcal{H}_K$. Fix constants $a, h > 0$, with $h \leq a$, sufficiently small that

$$a^2 + h^2 \leq R^2, \quad \bar{B}_X \sqrt{a^2 + h^2} \leq \frac{1}{2}, \quad 8C_\mu h^2 \leq \frac{1}{8}.$$

For an integer $M \geq 1$ and $\theta \in \{-1, 1\}^M$, define

$$W_\theta := a e_1 + \frac{h}{\sqrt{M}} \sum_{j=M+1}^{2M} \theta_{j-M} e_j, \quad g_\theta(x) := \langle W_\theta, \Phi(x) \rangle_{\mathcal{H}_K}.$$

Then

$$\|W_\theta\|_{\mathcal{H}_K} = \sqrt{a^2 + h^2} \leq R, \quad |g_\theta(x)| \leq \bar{B}_X \sqrt{a^2 + h^2} \leq \frac{1}{2}.$$

Put

$$A_M := \|g_\theta\|_{L^2(\nu)}^2 = a^2\mu_1 + \frac{h^2}{M} \sum_{j=M+1}^{2M} \mu_j.$$

This quantity does not depend on θ . Moreover,

$$a^2 c_\mu \leq A_M \leq \frac{1}{4}.$$

Step 2: construct admissible financial observations. Draw $x_t = (Z_t, r_t)$ independently with law ν .

Conditional on x_t , draw $s_t \in \{-1, 1\}$ according to

$$\mathbb{P}_\theta(s_t = 1 \mid x_t) = \frac{1 + g_\theta(x_t)}{2}.$$

The observed financial row is $(Z_t, R_{t+1}) = (Z_t, s_t r_t)$. For the least-favourable construction take

$$\mathcal{F}_t = \sigma(x_u, s_u : u < t) \vee \sigma(x_t).$$

This information structure is allowed by Definition 1. Rows are i.i.d. The managed feature is $s_t \Phi(x_t)$ and, since $s_t^2 = 1$,

$$\|s_t \Phi(x_t)\|_{\mathcal{H}_K}^2 = q_X.$$

Because x_t is contained in $\mathcal{F}_t$,

$$\mathbb{E}_\theta \left[\exp \left(\frac{\|s_t \Phi(x_t)\|_{\mathcal{H}_K}^2}{B_X^2} \right) \middle| \mathcal{F}_t \right] = \exp(q_X / B_X^2) \leq 2.$$

Moreover, for every $V \in \mathcal{H}_K$,

$$(R_{t+1}^p(V))^2 = \langle V, \Phi(x_t) \rangle_{\mathcal{H}_K}^2, \quad (R_{t+1}^p(V))^4 = \langle V, \Phi(x_t) \rangle_{\mathcal{H}_K}^4$$

conditionally on $\mathcal{F}_t$. Hence Condition (E2) holds with fourth-moment coefficient one, and therefore

with the class constant $K_4 \geq 1$. The second-moment operator of the managed feature is exactly Σ_0 .

For every $V \in \mathcal{H}_K$, conditional on the current x_t and the past,

$$\begin{aligned} \mathbb{E}_\theta \left[(1 - R_{t+1}^p(W_\theta)) R_{t+1}^p(V) \mid x_t, \mathcal{F}_{t-1} \right] \\ = \langle V, \Phi(x_t) \rangle_{\mathcal{H}_K} \mathbb{E}_\theta [s_t - g_\theta(x_t) \mid x_t, \mathcal{F}_{t-1}] = 0. \end{aligned}$$

Thus Condition (E4) holds. For any portfolio policy W with finite loss, write $g_W(x) = N^{-1} \sum_i W(z_i) r_i$. Then

$$Q_\theta(W) - Q_\theta(W_\theta) = \int (g_W(x) - g_\theta(x))^2 d\nu(x). \quad (\text{A.6})$$

Indeed, conditioning on x gives the quadratic $1 - 2g_\theta(x)g_W(x) + g_W(x)^2$. Consequently, W_θ is the population-optimal minimum-norm policy.

Its payoff satisfies

$$\mathbb{E}_\theta [R_{t+1}^p(W_\theta)] = A_M, \quad \mathbb{E}_\theta [(R_{t+1}^p(W_\theta))^2] = A_M.$$

Hence

$$(\text{SR}_\theta^\star)^2 = \frac{A_M}{1 - A_M}.$$

Because A_M is uniformly bounded away from zero and from one, the optimal Sharpe ratio is uniformly bounded away from zero and infinity. Thus the least-favourable family satisfies Condition (E6), with constants common to the family, as well as Conditions (E1)–(E5).

Step 3: neighboring economies are difficult to distinguish. Consider the more informative experiment that also reveals x_t and s_t . A lower bound in this augmented experiment also applies to the financial observations. Neighboring parameters differ in one sign, say coordinate $j - M$. Their conditional Bernoulli probabilities lie in $[1/4, 3/4]$, and

$$\text{KL}(\text{Ber}(p), \text{Ber}(q)) \leq \frac{(p - q)^2}{q(1 - q)}.$$

Hence, for the laws $\mathbb{P}_\theta^{\text{aug}, T}$ of T augmented observations and a one-bit neighbor θ' , we have

$$\begin{aligned} \text{KL}(\mathbb{P}_\theta^{\text{aug}, T}, \mathbb{P}_{\theta'}^{\text{aug}, T}) &\leq 2T \int (g_\theta - g_{\theta'})^2 d\nu \\ &= \frac{8Th^2\mu_j}{M} \\ &\leq 8C_\mu h^2 \frac{T}{M^{b+1}}. \end{aligned}$$

Choose

$$M = \lceil T^{1/(b+1)} \rceil.$$

The right-hand side is at most $1/8$, so Pinsker's inequality gives total variation at most $1/4$ for every neighboring pair.

Step 4: testing difficulty implies scale-free directional error. Define

$$\psi_j(x) := \frac{\langle e_j, \Phi(x) \rangle_{\mathcal{H}_K}}{\sqrt{\mu_j}},$$

so that $(\psi_j)_{j \geq 1}$ is orthonormal in $L^2(\nu)$. Let $\widehat{g}$ be any estimated square-integrable payoff rule and write

$$\widehat{a}_j := \langle \widehat{g}, \psi_j \rangle_{L^2(\nu)}.$$

Decode the signs by

$$\widehat{\theta}_{j-M} := \text{sign}(\widehat{a}_1 \widehat{a}_j), \quad M < j \leq 2M,$$

with any fixed convention at zero. This decoder is invariant to any nonzero global rescaling of $\widehat{g}$.

Define the scale-free directional error

$$D_\theta(\widehat{g}) := \inf_{c \in \mathbb{R}} \|g_\theta - c\widehat{g}\|_{L^2(\nu)}^2.$$

For any $c \in \mathbb{R}$, if $c\widehat{a}_1 \leq 0$, the squared error in the first coordinate is at least $a^2\mu_1$. If $c\widehat{a}_1 > 0$,

every incorrectly decoded sign contributes at least $h^2\mu_j/M$ to the squared error. Since $h \leq a$ and $H(\widehat{\theta}, \theta) \leq M$, both cases imply

$$D_\theta(\widehat{g}) \geq \frac{h^2 c_\mu}{2^b M^{b+1}} H(\widehat{\theta}, \theta).$$

For every neighboring pair, the sum of the two sign-error probabilities is at least $1 - \text{TV} \geq 3/4$. Averaging over the hypercube and summing over its M coordinates therefore gives

$$2^{-M} \sum_{\theta} \mathbb{E}_{\theta}[H(\widehat{\theta}, \theta)] \geq \frac{3M}{8}.$$

Consequently,

$$2^{-M} \sum_{\theta} \mathbb{E}_{\theta}[D_\theta(\widehat{g})] \geq \frac{3h^2 c_\mu}{2^{b+3}} M^{-b}.$$

Step 5: directional error is a Sharpe-ratio gap. Write

$$m := \langle g_\theta, \widehat{g} \rangle_{L^2(\nu)}, \quad \nu := \|\widehat{g}\|_{L^2(\nu)}^2.$$

If $\nu > 0$, the payoff generated by $\widehat{g}$ satisfies

$$\mathbb{E}_{\theta}[R(\widehat{g})] = m, \quad \mathbb{E}_{\theta}[R(\widehat{g})^2] = \nu,$$

and hence

$$\text{SR}_{\theta}(\widehat{g})^2 = \frac{m^2}{\nu - m^2}.$$

Moreover,

$$D_{\theta}(\widehat{g}) = A_M - \frac{m^2}{\nu} = \frac{1}{1 + \text{SR}_{\theta}(\widehat{g})^2} - \frac{1}{1 + (\text{SR}_{\theta}^*)^2}.$$

The same identity holds for $\nu = 0$ under the convention $\text{SR}_{\theta}(\widehat{g}) = 0$. Cauchy–Schwarz implies $|\text{SR}_{\theta}(\widehat{g})| \leq \text{SR}_{\theta}^*$.

If $\text{SR}_\theta(\widehat{g}) \geq 0$, then

$$\text{SR}_\theta^\star - \text{SR}_\theta(\widehat{g}) \geq \frac{D_\theta(\widehat{g})}{2\bar{S}}.$$

If $\text{SR}_\theta(\widehat{g}) < 0$, then the gap is at least $\underline{S}$; since $D_\theta(\widehat{g}) \leq A_M \leq 1/4$, it is again bounded below by a fixed positive multiple of $D_\theta(\widehat{g})$. Therefore there is a constant $c_S > 0$, independent of M, T, θ , such that

$$\text{SR}_\theta^\star - \text{SR}_\theta(\widehat{g}) \geq c_S D_\theta(\widehat{g}).$$

Combining this inequality with Step 4 yields

$$2^{-M} \sum_{\theta} \mathbb{E}_{\theta} [\text{SR}_\theta^\star - \text{SR}_\theta(\widehat{g})] \geq c_1 M^{-b}$$

for a constant $c_1 > 0$. Since $M = \lceil T^{1/(b+1)} \rceil \leq 2T^{1/(b+1)}$ for all sufficiently large T , we obtain

$$\sup_{\theta} \mathbb{E}_{\theta} [\text{SR}_\theta^\star - \text{SR}_\theta(\widehat{g})] \geq c_2 T^{-b/(b+1)}$$

for some $c_2 > 0$.

The argument allowed $\widehat{g}$ to range over all square-integrable estimators in the more informative augmented experiment, a larger class than managed portfolio rules based on the observed financial rows. Taking the infimum over all measurable portfolio-learning procedures proves

$$\inf_{\widetilde{W}_T} \sup_{\varepsilon \in \mathcal{E}} \mathbb{E}_{\varepsilon} \left[\text{SR}_{\varepsilon, K}^\star - \text{SR}_{\varepsilon} \left(R_{t+1}^p(\widetilde{W}_T) \right) \right] \geq c T^{-b/(b+1)}$$

for a constant $c > 0$ and all sufficiently large T . □

B. Tabelle

Table VII. Portfolio representations generated by different kernels

Representation	Kernel	Portfolio policy
Common exposure	$K(z, z') = 1$	$W(z)$ is constant
Linear	$K(z, z') = 1 + z'z'$	$W(z)$ is linear in characteristics
Finite features	$K(z, z') = \Phi(z)' \Phi(z')$	$W(z)$ lies in the span of the chosen features
Polynomial	$K(z, z') = (1 + z'z')^q$	$W(z)$ includes polynomial terms and interactions
Gaussian	$K(z, z') = \exp\left(-\frac{\ z - z'\ ^2}{2\ell^2}\right)$	Flexible nonlinear $W(z)$
Matérn	$K_s(z, z') = c_{s,D} \rho_s(z, z')^{s-D/2} \times \mathcal{K}_{s-D/2}(\rho_s(z, z'))$	Flexible nonlinear $W(z)$ with smoothness controlled by s
Neural tangent	$K_{\text{NTK}}(z, z') = \frac{\ z\ \ z'\ }{2\pi} \left[\sin \theta + 2(\pi - \theta) \cos \theta \right]$	Infinite-width one-hidden-layer ReLU policy

Notes: Here $\ell > 0$ denotes the kernel length scale. For the Matérn kernel,

$$\rho_s(z, z') = \sqrt{2s - D} \frac{\|z - z'\|}{\ell}, \quad c_{s,D} = \frac{2^{1-(s-D/2)}}{\Gamma(s - D/2)}, \quad s > \frac{D}{2},$$

and $\mathcal{K}_\alpha$ denotes the modified Bessel function of the second kind. Under standard regularity conditions, the Matérn RKHS is equivalent to a Sobolev space of smoothness s .

For the neural tangent kernel,

$$\theta = \arccos\left(\frac{z'z'}{\|z\| \|z'\|}\right),$$

and the displayed expression corresponds to a one-hidden-layer ReLU network without bias in the infinite-width NTK limit, up to the conventional overall normalization.

The kernel determines the class of characteristic-to-exposure mappings available to the portfolio manager. Portfolio returns use the common cross-sectional normalization in Equation (1). The common-exposure specification is an equal-exposure characteristic rule, not an unrestricted vector of static stock weights.

Table VIII relates the learning exponent to familiar spectral benchmarks. These are conditions on the managed-portfolio operator, not conclusions implied by the kernel name alone.

Table VIII. Benchmark portfolio-learning rates

Portfolio class	Managed-portfolio spectrum	Sharpe learning rate
Finite dimensional	Fixed finite rank	T^{-1}
Gaussian benchmark	Fast decay	Nearly T^{-1}
Sobolev–Matérn benchmark	$b = 2\tilde{s}/D$	$T^{-2\tilde{s}/(2\tilde{s}+D)}$

Notes: Sharpe rates use Condition (E6). The finite-rank rate uses a fixed identifiable payoff space and root- T moment estimation. The fast-decay row is a separate i.i.d. or sufficiently fast-mixing benchmark, with possible logarithmic factors and an admissible regularization sequence; it does not follow by setting $b = \infty$ in Theorem 1. The Sobolev–Matérn row assumes the displayed *managed-payoff* spectral law and $\gamma > b$.

Simulation appendix

Population evaluation uses independent group quadrature. In orthonormal Nyström coordinates $\phi(z)$, write $H_g = \frac{1}{3} \sum_r \phi(Z_{g,r})\beta(Z_{g,r})'$ and $\bar{H} = M^{-1} \sum_g H_g$. Then

$$\hat{\Sigma} = \frac{\sigma_\varepsilon^2}{N} \hat{T}_K + \frac{1}{GM} \sum_g H_g H_g' + \left(1 - \frac{1}{G}\right) \bar{H} \bar{H}', \quad \hat{T}_K = \frac{1}{3M} \sum_{g,r} \phi(Z_{g,r}) \phi(Z_{g,r})'.$$

This integrates the factor and idiosyncratic shocks analytically while preserving within-triplet dependence. The quadrature distribution is independent of all training paths and cannot affect validation choices. For empirical ranks, full independent cross-sections are integrated because ranking couples groups. For heteroskedasticity, the idiosyncratic term uses the actual bounded state-dependent conditional variance.

The headline population quadrature estimates the regret floor of the unregularized rank-256 subspace at 0.00002098. This approximation floor is included in every displayed population-bias curve.

The paired numerical-rank check uses the first 200 headline economic paths. Increasing rank from 256 to 512 changes mean oracle or selected regret by at most 4.64 percent across all training lengths, against the predeclared 10 percent tolerance. The lower rank 128 is reported as a separate sensitivity. For the first 20 fixed headline policies, increasing independent population quadrature from 8,192 to 32,768 groups changes mean evaluated regret by at most 1.58 percent (5 percent tolerance). This latter comparison holds fitted policies and penalties fixed; it does not reselect tuning parameters.

Table IX. Spectral and learning-rate sensitivity across Matérn smoothness values. Fitted learning exponents use the full predeclared T grid.

Kernel	Spectral decay		Regret exponent		Complexity exponent	
	Theory	Fitted	Theory	Fitted	Theory	Fitted
$\nu = 1.5$	1.5	1.24	0.6	0.5177	0.4	1.441
$\nu = 0.5$	1.167	1.004	0.5385	0.498	0.4615	1.807
$\nu = 2.5$	1.833	1.365	0.6471	0.5656	0.3529	1.564

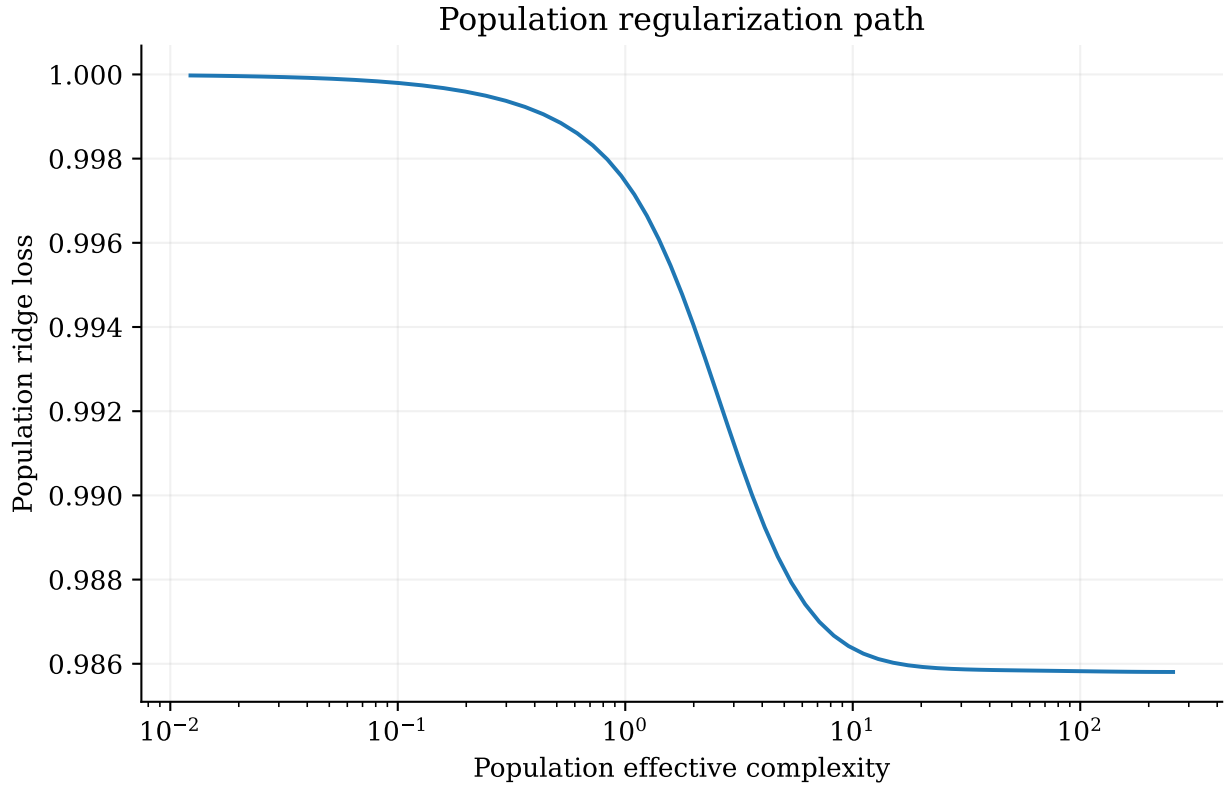


Figure 17. Population ridge loss versus population effective complexity.

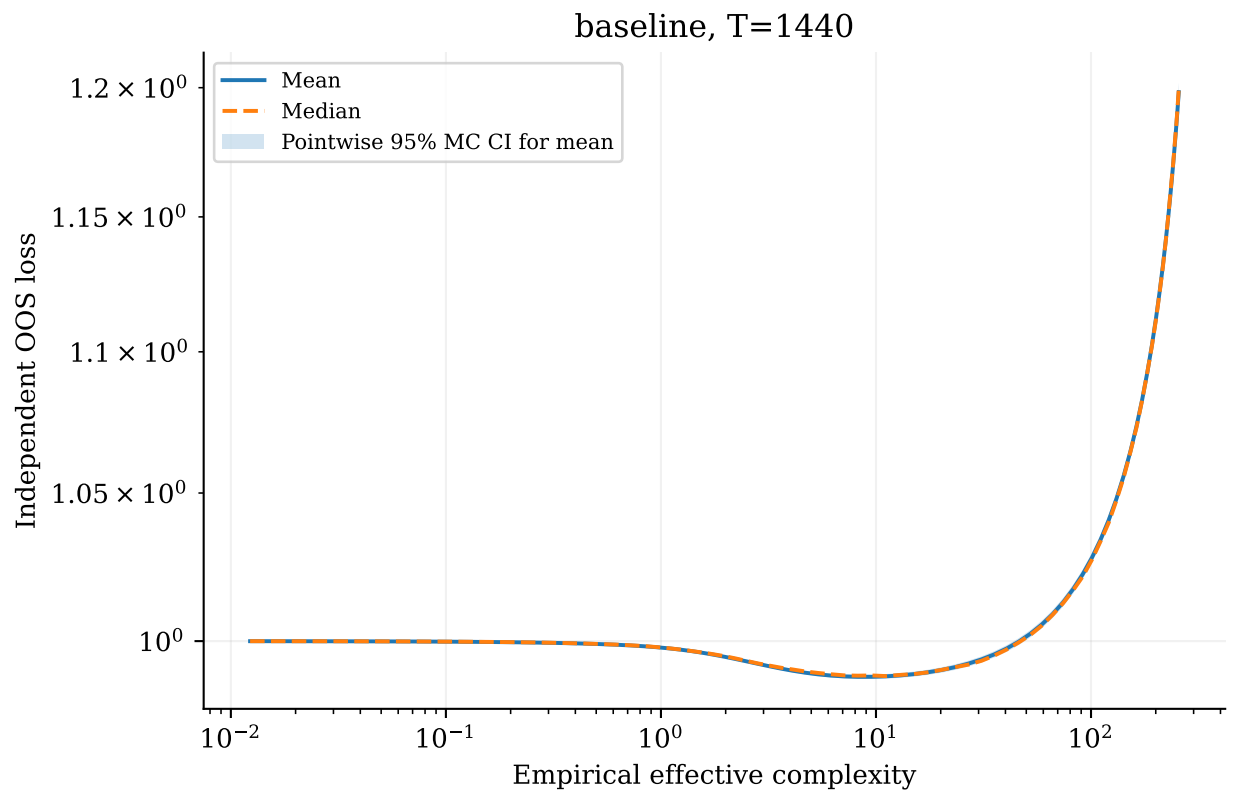


Figure 18. OOS loss: Monte Carlo mean, median, and pointwise confidence interval for the mean.

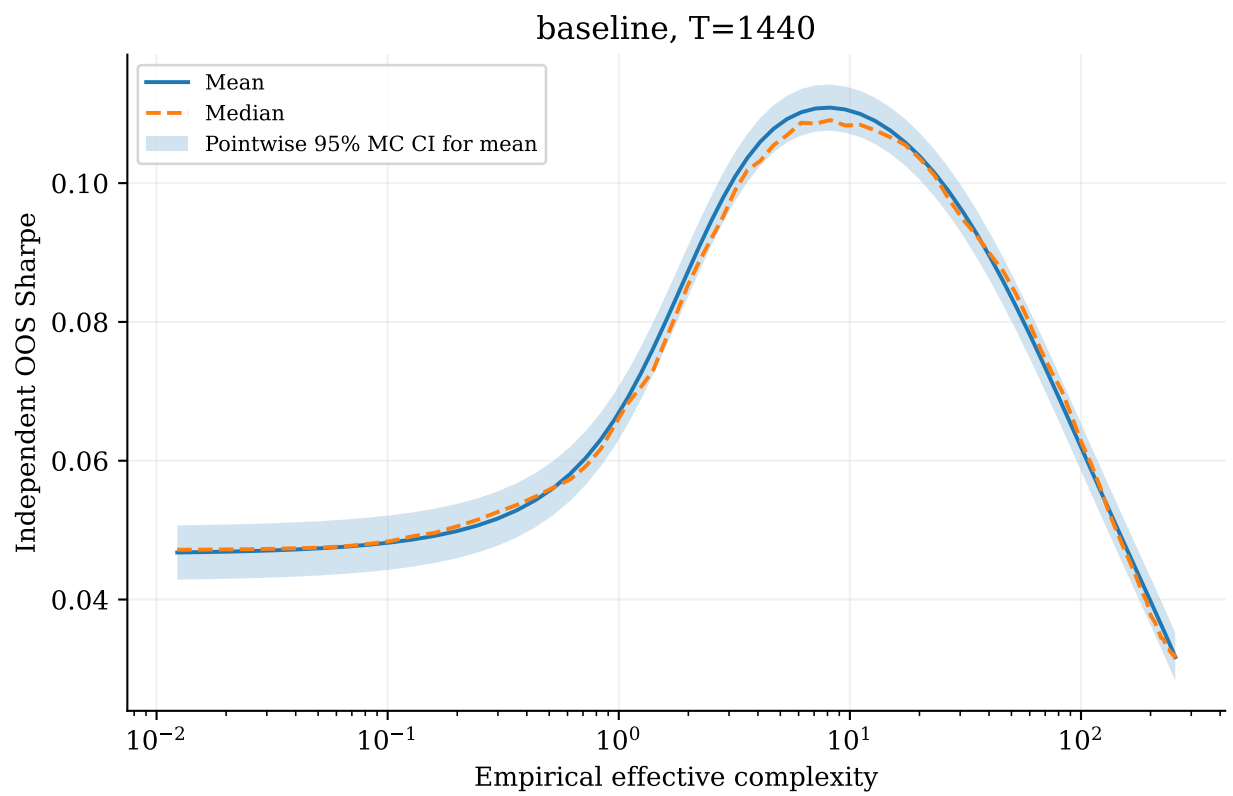


Figure 19. OOS Sharpe: Monte Carlo mean, median, and pointwise confidence interval for the mean.

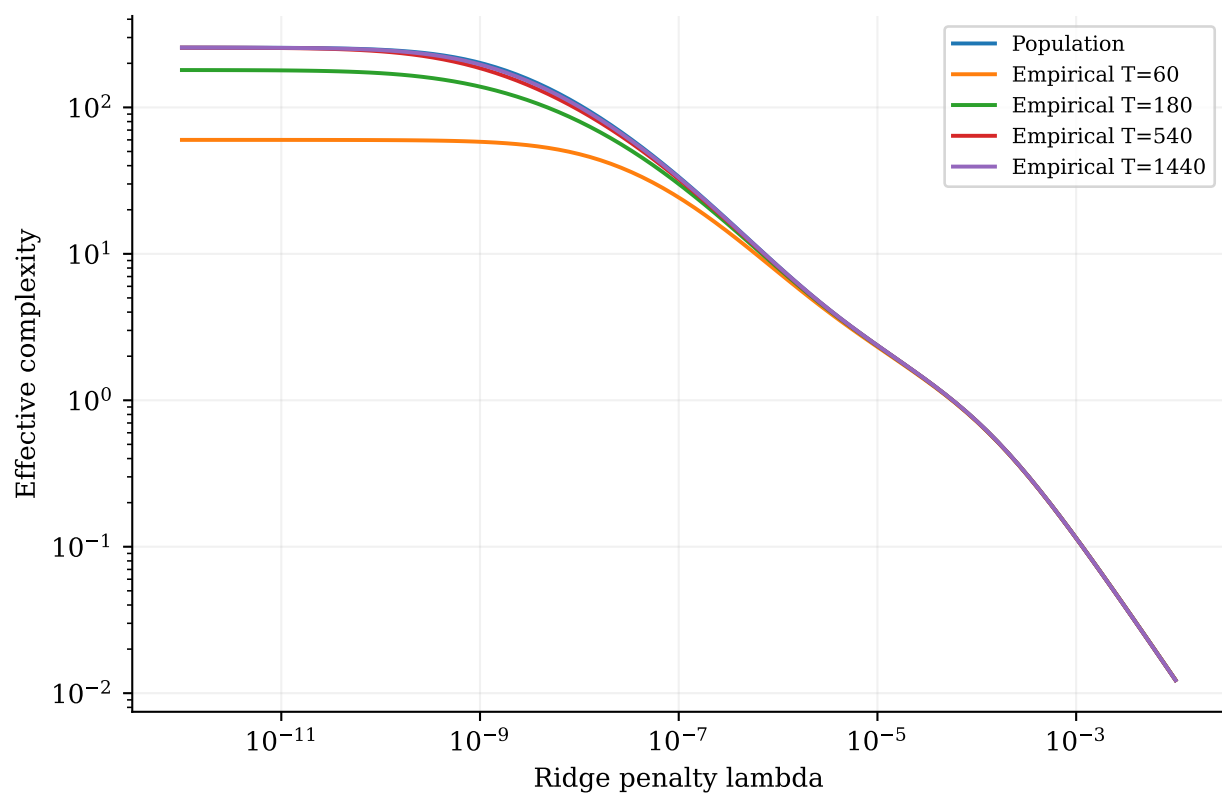


Figure 20. Population and empirical effective complexity as reparameterizations of regularization.

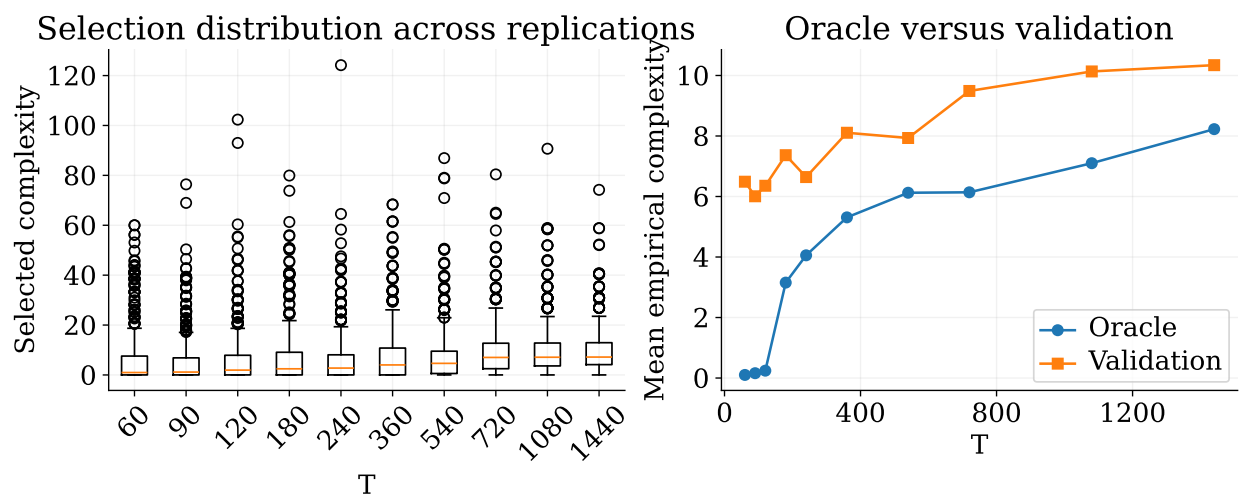


Figure 21. Selection distributions and oracle versus selected empirical complexity.

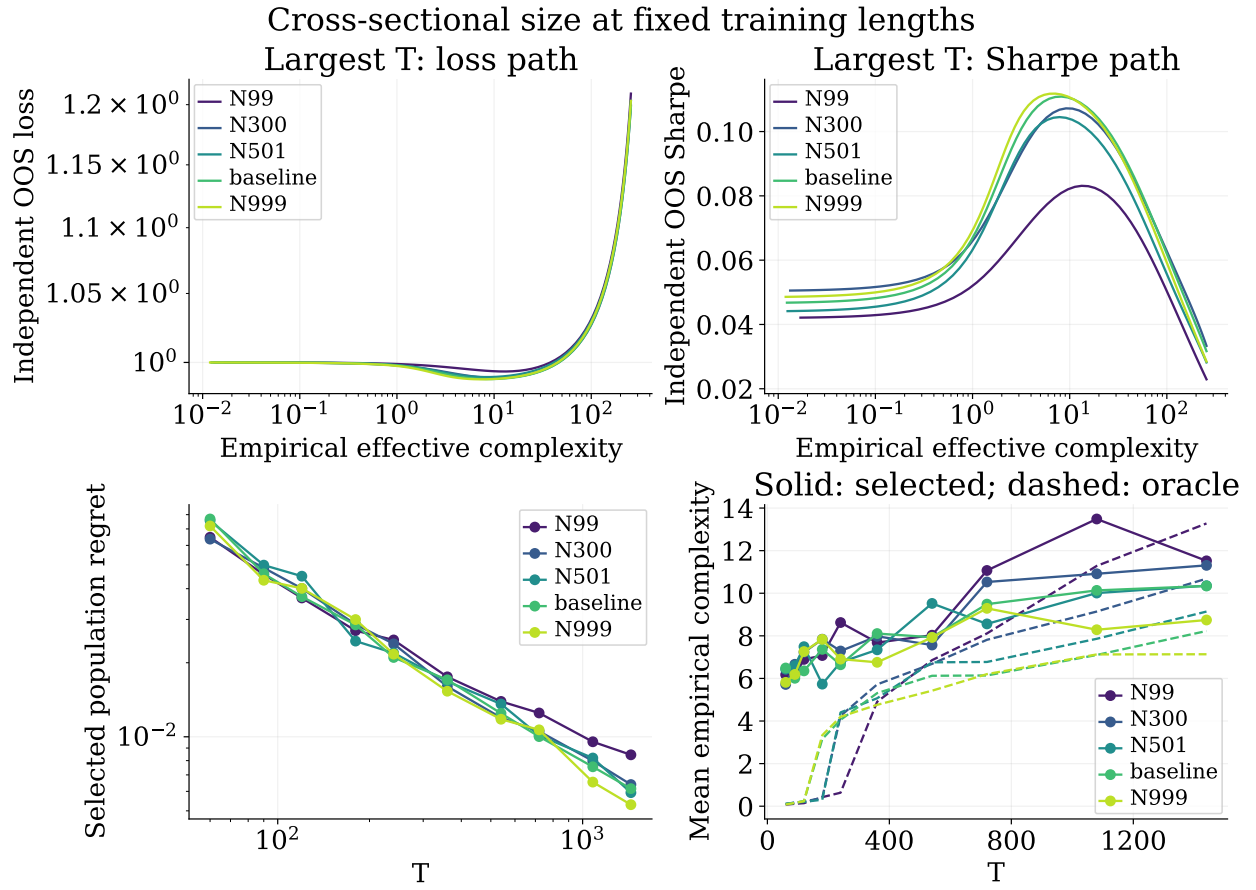


Figure 22. Cross-sectional-size robustness at every fixed training length.

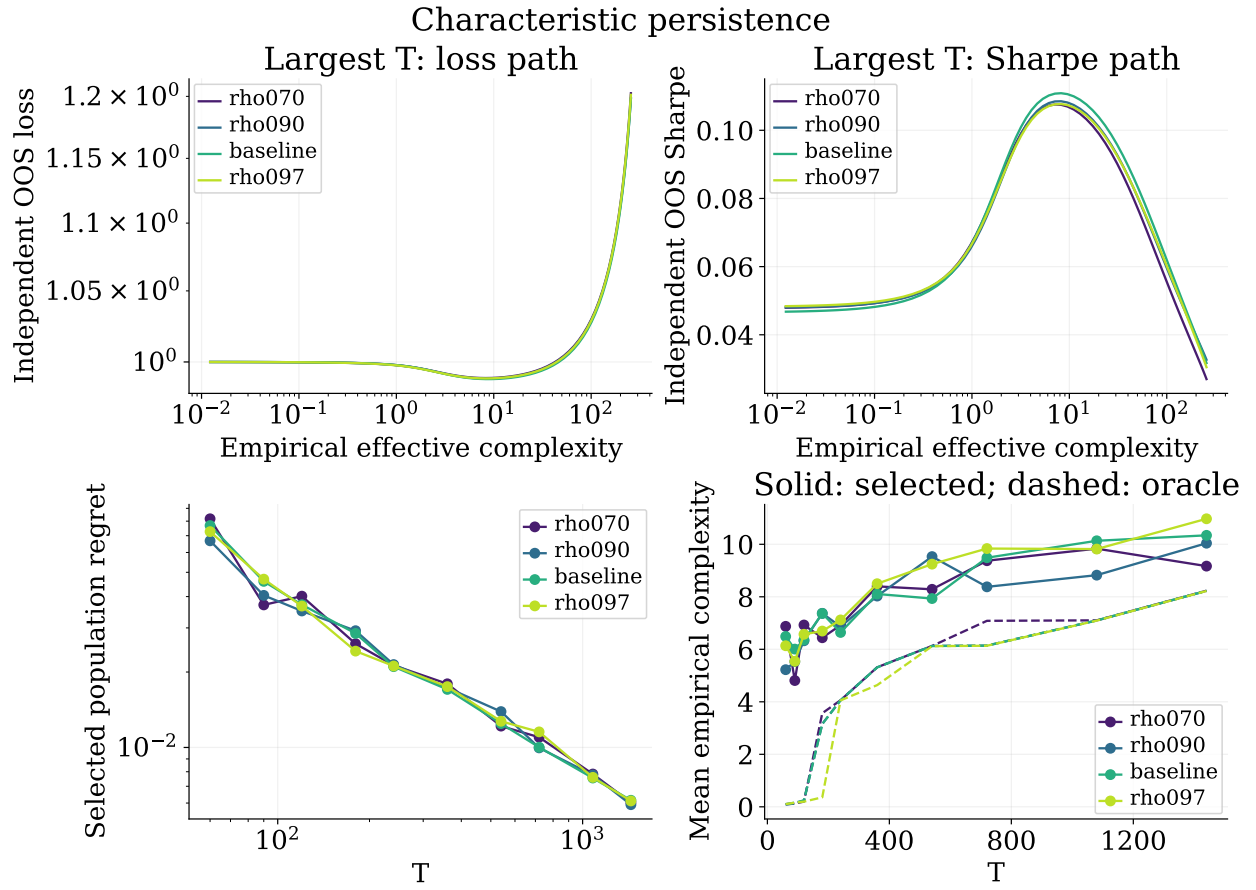


Figure 23. Persistence robustness.

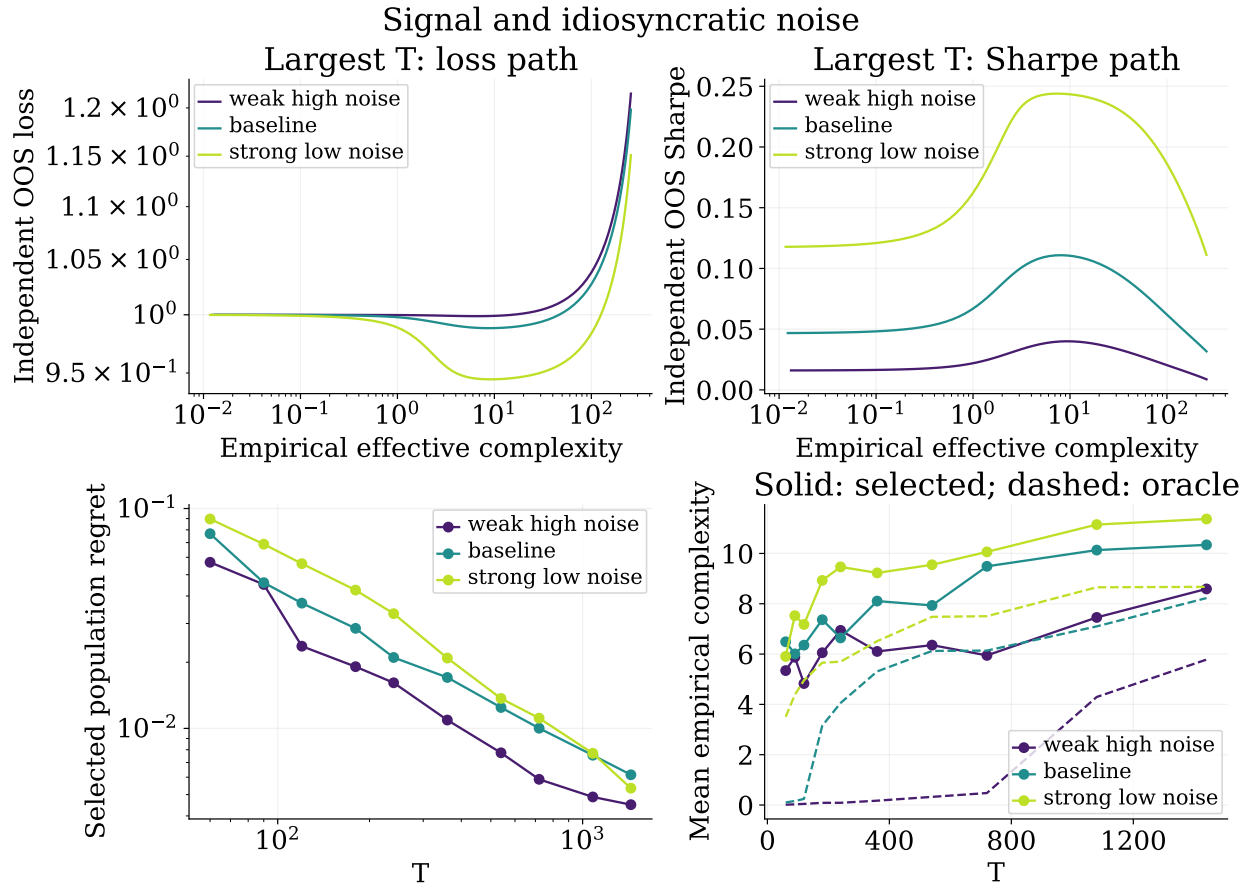


Figure 24. Signal-to-noise robustness.

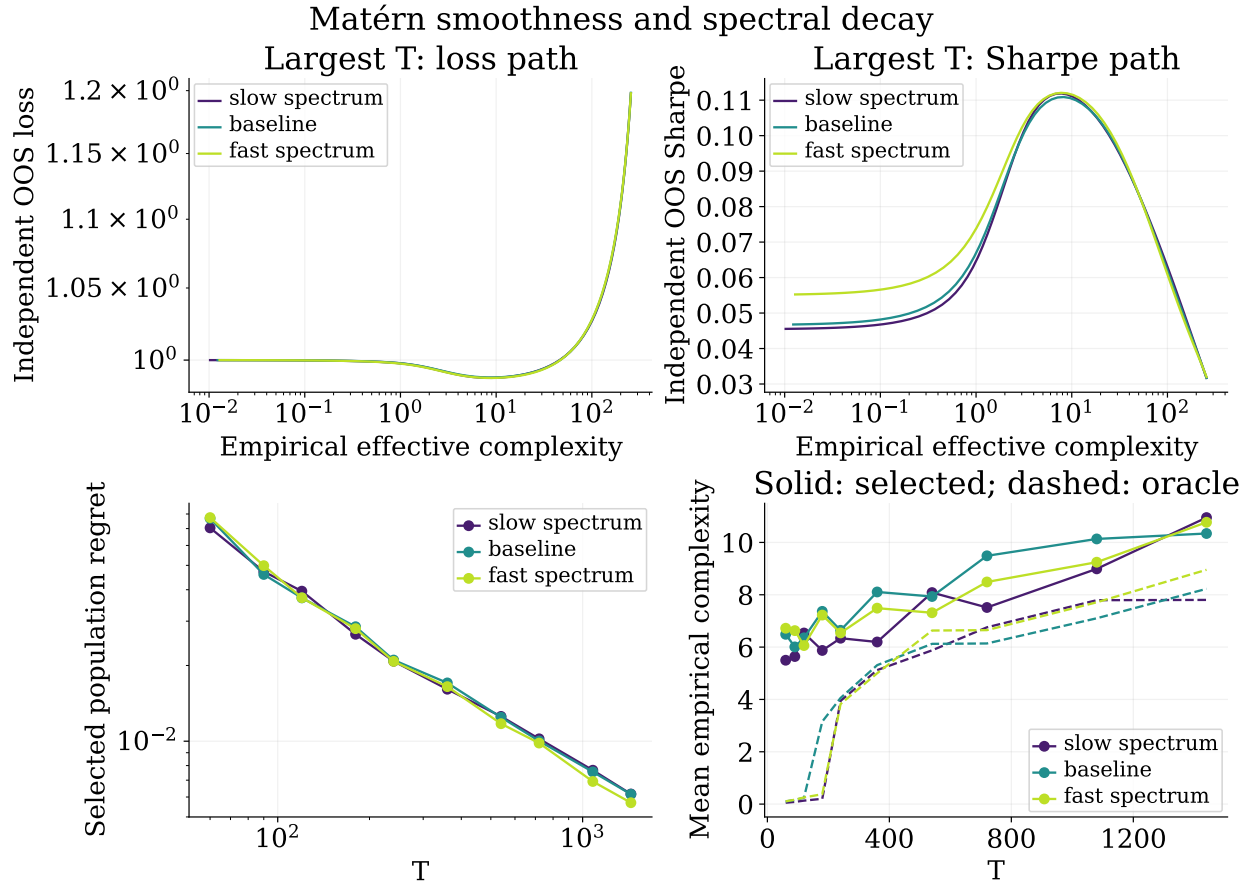


Figure 25. Spectral-decay robustness across Matérn smoothness values.

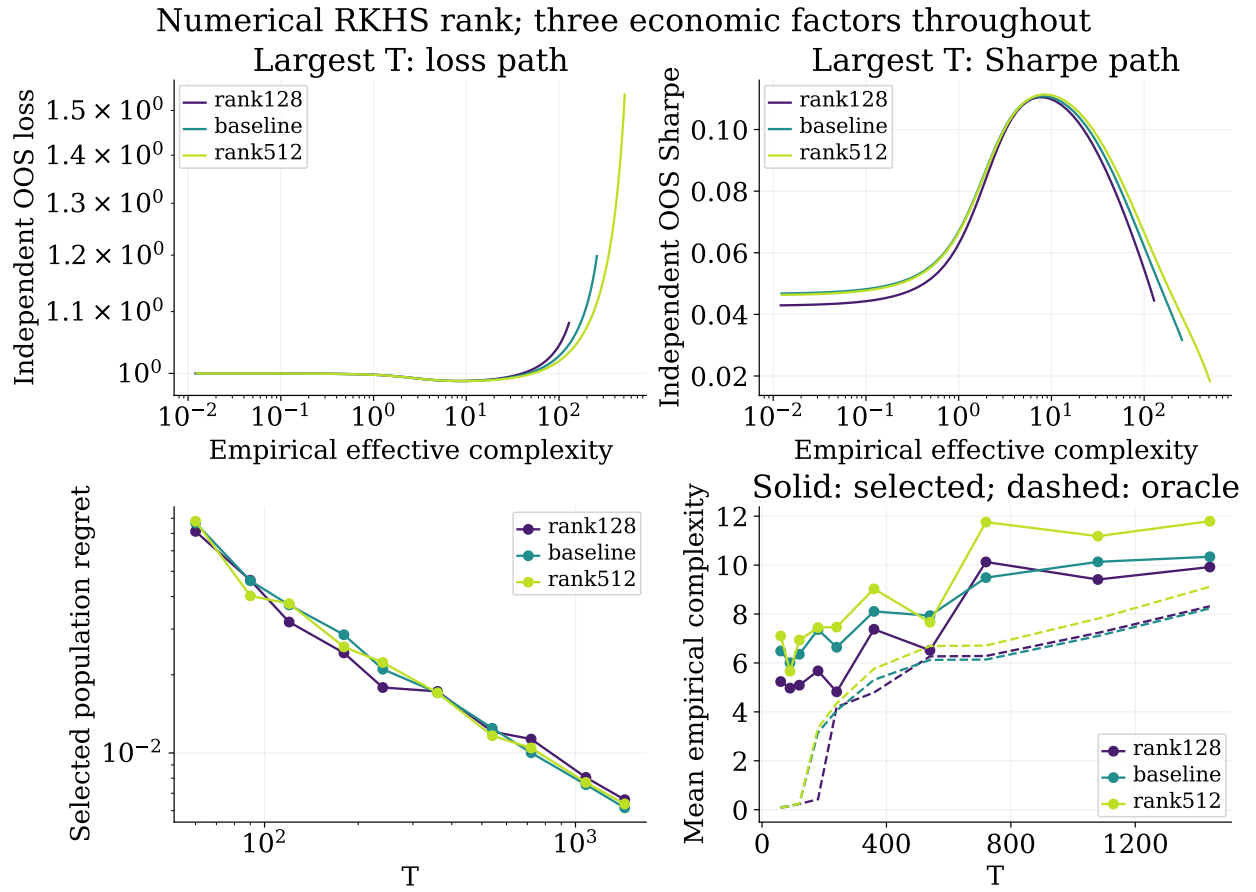


Figure 26. Numerical approximation-rank sensitivity; the economic factor dimension remains three.

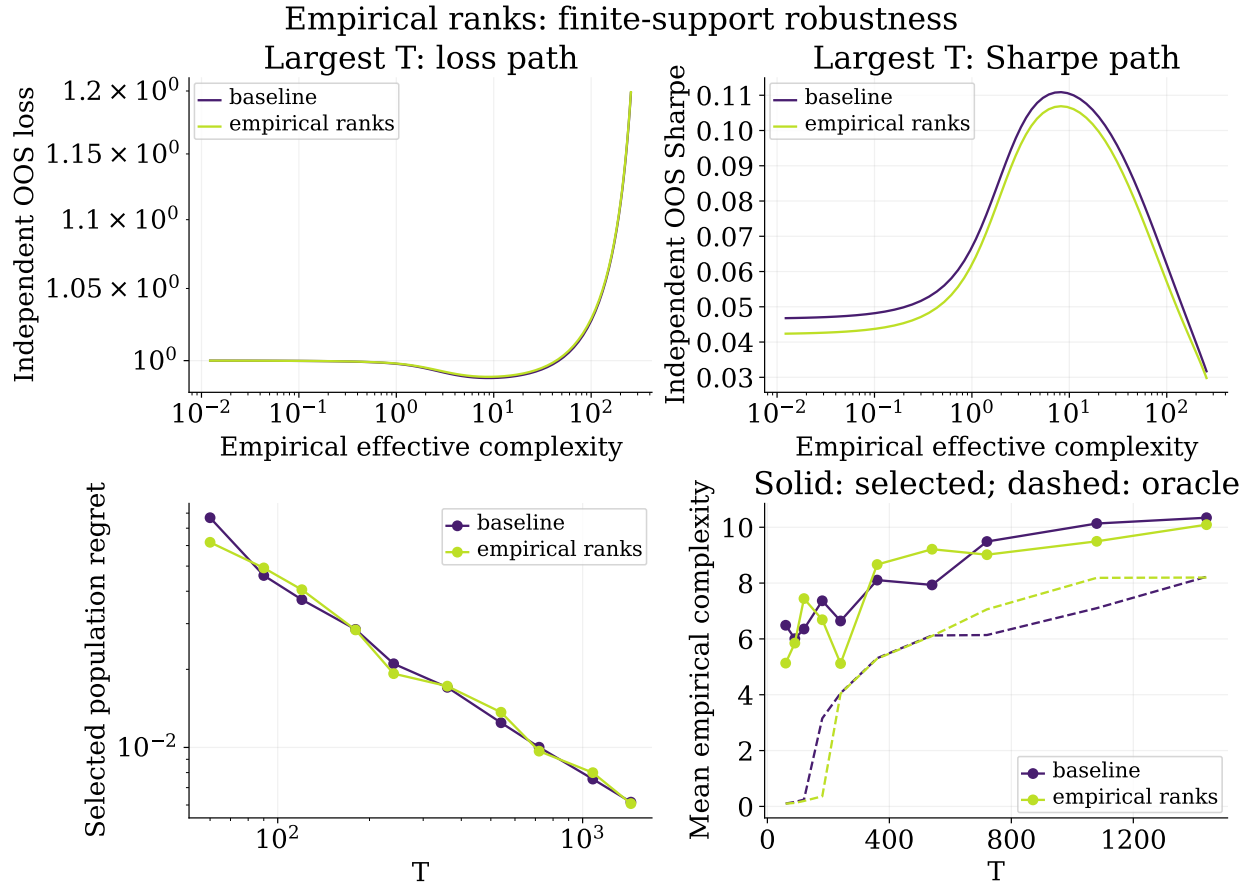


Figure 27. Empirical-rank robustness: E5 has finite support and does not have the baseline asymptotic interpretation.

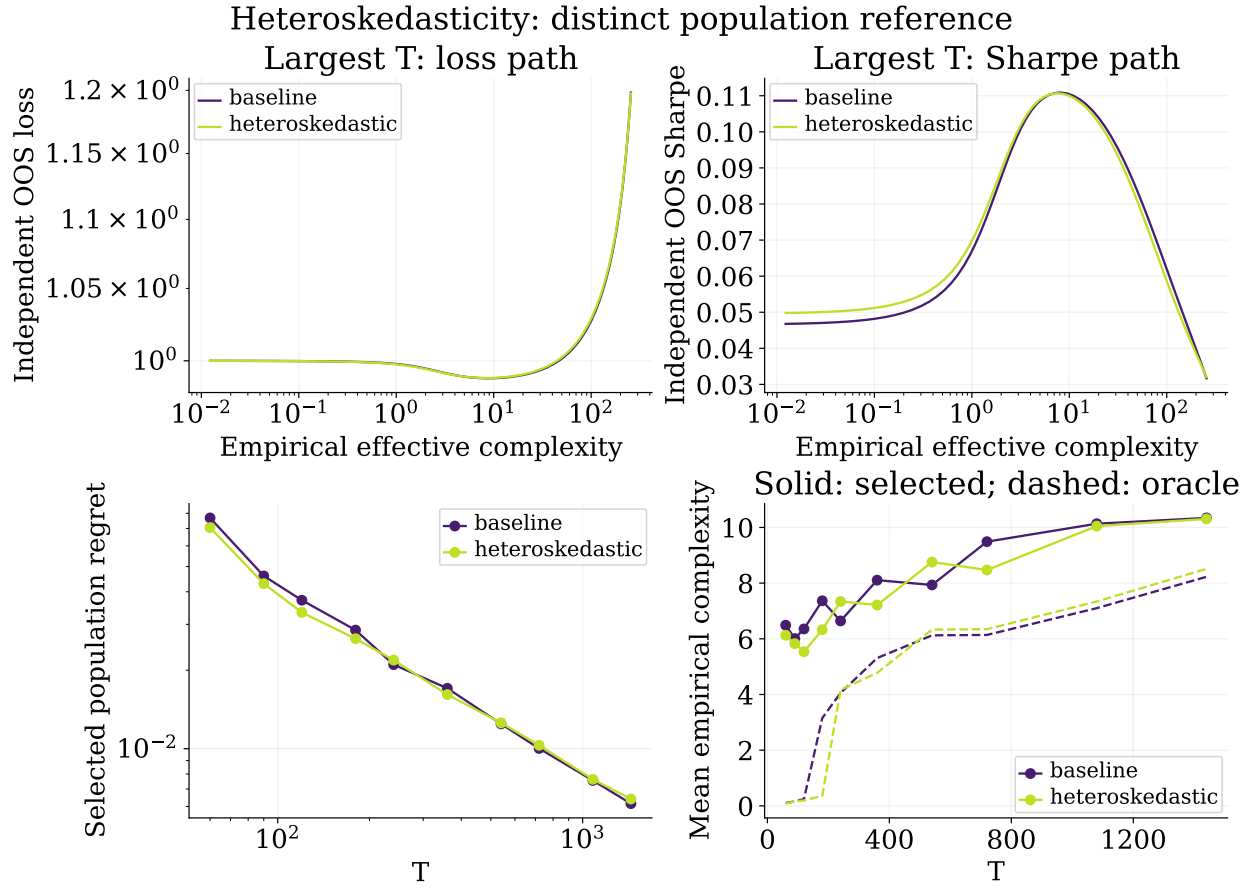


Figure 28. Bounded heteroskedasticity: conditional pricing is not imposed and regret uses a finite-subspace reference.